Export Destinations, Quality, and Automation: Evidence from China *†

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Abstract

Automation technology not only saves labor costs but also facilitates quality upgrading, so growing export demands in high-income destinations with stronger quality preference may induce exporters to automate more actively. We illustrate this insight in a model with non-homothetic preference and comparative advantage in quality production generated by automation technology. We test the key model prediction using Chinese firm-level trade data from 2001 to 2015, and find that increased demand from high-income destinations causes rising firm-level imports of automation machines, while increased demand from low-income destinations does not. This effect is driven by the imports of regulating and control instruments, a type of automation machines crucial for quality control and monitoring, and by imports sourced from high-income economies. Our finding highlights the importance of access to high-income destinations for promoting technology adoption in developing economies.

Keywords: Automation, Non-homotheticity, Demand shocks

JEL Classifications: F10; F14; F19

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1 Introduction

Export is regarded as an important driving force of firm technology adoption. A growing export market induces technology adoption that entails fixed-cost investment, a "scale effect" that has been examined by existing studies (e.g., Lileeva and Trefler, 2010; Bustos, 2011). We complement the existing literature by showing that, in addition to the "scale effect", export destinations also matter for technology adoption. Since customers in richer destinations prefer higher-quality goods, increasing exposure to richer destinations encourages exporting firms to not only upgrade their skill compositions and input quality (e.g., Brambilla et al., 2012; Bastos et al., 2018), but also adopt technology that facilitates quality upgrading.

In this paper, we find that ....., an important technology that becomes increasingly adopted in manufacturing during the last few decades. While the roles of automation as labor-saving and labor-augmenting technologies are extensively investigated, its capability in improving and maintaining product quality is relatively understudied.[go to literature] Industrial experience shows that automation equipment increases the precision, consistency and traceability of the manufacturing and quality control processes. Examples include optical measuring instruments that inspect key car components (e.g., bodies in white; battery casings) with high precision and reduce defect rates from 120 per million to 5 per million, and automatic regulating systems that enhance boiler operation stability and prevent quality issues due to production fluctuations. [go to R&C illustration] DeStefano and Timmis (2024) also offer detailed discussions on how the use of industrial robots facilitates product quality upgrading. Given

of, the motive to upgrade product quality may also cause automation. In this paper, we exploit demand shocks from high-income export markets to investigate the extent to which increased demand for high-quality goods induces automation

among Chinese manufacturing firms.

Automation in manufacturing has increased remarkably across the globe over the past few decades, with China standing out as one of the most active adopters.[1]

[1] For example, based on BACI-CEPII data, in 2022, China accounts for 12.2% of the world imports of automation machines and 19.5% of the world imports of industrial robots, ranking the 2nd and the 1st in all economies respectively.

In the case of China, which has become a major exporter to advanced economies over the past decades, rising demand for high-quality goods in these markets has likely created additional incentives for firms to automate.

As pointed out by Verhoogen (2023), the motive of quality upgrading is particular relevant for exporting firms in developing economies, who usually face richer foreign customers in the export market than their domestic customers,

Leveraging the machine classification developed by Acemoglu and Restrepo (2022), we first show that imported automation machines have increased significantly relative to other types of machines for industrial production in China during 2000-2015. Moreover, we uncover the dominant role of the subgroup "regulating and control instruments", which mainly includes measuring, controlling and monitoring instruments and appliances and hence plays a crucial role in quality production. In China, the share of regulation and control instruments in the import value of automation machines has grown from 46.4% in 2001 to 65.0% in 2015, and the nominal value of imported regulating and control instruments has increased more than 16 times. We also find similar magnitudes of importance of regulating and control instrument in the world import of automation machines. The imports of industrial robots have also achieved remarkable growth, but only account for no larger than 2.3% of China's import value of automation machines during the sample period. Therefore, the fast-growing imports of industrial robots remain a small subgroup in the whole category of automation machines. We thus offer new descriptive evidence regarding a specific type of automation machines, which have not yet received sufficient attention in the previous studies but are potentially critical inputs for quality upgrading in the manufacturing sector.

We then propose a model of automation adoption to shed light on how demand shocks in different markets may affect the incentive to automate. Firms face non-homothetic consumers in different markets of different income levels, and high-income destinations have stronger preference for high-quality goods. Firms decide the prices and quality of their goods served to each market and their automation levels. The marginal benefit of quality upgrading increases as high-income destinations see growing demands. We describe an increase in firm's automation level as a costly decision to raise the firm's comparative advantage in producing high-quality goods, as automation machines increase precision and accuracy in the production process. We formulate this comparative advantage feature in

production by building on Fieler et al. (2018) and Sun (2019), who assume that high-quality input and advanced machines generate comparative advantages in producing high-quality output, respectively. Our model hence generates novel testable predictions about how a firm adjusts its automation level in responses to different demand shocks: a firm has a stronger (weaker) incentive to increase its automation level when facing positive demand shocks in high-income (low-income) markets.

We use detailed Chinese Customs Data that covers transactions of every goods imported or exported by all firms to test the model prediction regarding the impacts of high-income demand shocks on automation. We measure firm-level automation using firm-level imports of automation machines as identified by the classification of Acemoglu and Restrepo (2022). To identify the causal effects of export demand shocks in different markets, we construct firm-specific export demand shocks in high-income destinations and low-income destinations, respectively. We leverage the exogeneity of demand shocks at the product-destination level to achieve identification. We test the difference between the estimated coefficients of high-income and low-income destinations.

We find that a 10% increase in demand from high-income export destinations causes a 2.1% increase in firm-level imports of automation machines. In contrast, the same increase in demand from low-income export destinations yields no significant effect. Importantly, the difference between the effects of demands from high-income and low-income destinations is statistically significant. Notably, this result is mainly driven by the imports of "regulating and control instruments", precisely the equipment used to increase the precision and reliability of production. Finally, we also show that high-income demand shocks lead to increases in export prices and export quality, consistent with the motive to upgrade quality by automating. As placebo tests, we also show that our main result only holds for manufacturing firms and the imports of automation machines, but not for trade intermediaries or imports of other types of machines.

We also find that demand shocks from high-income destinations have stronger effects in more differentiated sectors, in industries with greater price dispersion, and among initially larger firms and foreign-invested firms. These heterogeneous responses underscore the role of firm scale, managerial capability, and sectoral characteristics in shaping firms' adaptability to global demand shocks. Sectors with higher product differentiation or price dispersion

tend to reap greater gains from automation, as quality upgrading yields higher returns in these environments. Likewise, larger or foreign-invested firms are likely to have superior managerial capability, allowing them to respond more effectively to high-income market demand through investment in automation.

This paper aims to contribute to three strands of literature. The first strand studies the effect of market size on technological adoption and innovation. Existing literature suggests a positive "market size" effect of technology adoption and innovation, with larger markets leading to greater incentives for firms to invest in new technologies (Lileeva and Trefler, 2010; Bustos, 2011; Aghion et al., 2024). Building on this literature, we highlight increased automation as another form of technology adoption, and take a further step by showing that the "market size" effect of such a technology adoption may be stronger for high-income destinations. The second strand focuses on the impact of export on firm upgrading. Previous studies have shown that increased high-income demand shocks induce skill upgrading (Brambilla et al., 2012; Demir et al., 2024) and input upgrading (Bastos et al., 2018), and have emphasized the role of access to market (Atkin et al., 2017b). One novelty of this paper is to show that the need to upgrade quality induces automation, suggesting that automation is a crucial mechanism for quality upgrading. Importantly, we also show that the increased imports of automation machines are mainly driven by "regulating and control instruments", a type of equipment critical for quality monitoring and quality control. Therefore, our finding also complements DeStefano and Timmis (2024), who find that increased usage of industrial robots enhances export quality. The third strand studies the driving forces for automation adoption. Existing literature emphasizes the role of rising labor costs (Fan et al., 2021; Acemoglu and Restrepo, 2022), organizational barriers (Atkin et al., 2017a), and competition (Bastos et al., 2023). Our focus on the role of demands in high-income destinations in driving automation is distinct in the literature.

The rest of the paper is organized as follows. Section 2 describes the data and the imports of automation machines in China. Section 3 presents the theoretical model and its key predictions. Section 4 lays out the empirical strategy. Section 5 reports empirical findings, and Section 6 concludes.

2 Imports of Automation Machines and the Role of Regulating and Control Instruments

This section introduces the data we used in this paper and analyzes the patterns on imports of automation equipment. In particular, we document new descriptive evidence highlighting the role of "regulating and control instruments" in driving the growth of imported automation machines, a subcategory supposed to enhance manufacturers' capacity of quality production.

2.1 Data

We draw on two main datasets to conduct the analysis in this paper: the Chinese Customs Data and the Base pour l'Analyse du Commerce International (BACI henceforth) data, both from 2000 to 2015.

The Chinese Customs Data report export and import value and quantity by product, classified by 8-digit Harmonized System (HS) code, between Chinese firms and their trading partner countries via various trade modes. It hence provides comprehensive information on international trade at the firm-product-country-year level. Importantly, using the information of firm names in Chinese in the dataset, we can distinguish trade intermediaries from manufacturing firms following the approach developed by Ahn et al. (2011) and Poncet and Xu (2018). The subsequent analysis focuses on international trade conducted by manufacturing firms in their ordinary trade activities (so processing trade is excluded), to reflect the autonomous business decisions made by manufacturers.

The BACI data provides yearly data on bilateral trade flows across more than 200 trading partner countries at the product level. Products are identified using the Harmonized System (HS), which is the standard nomenclature for international trade. We use BACI Dataset (202401b Version) and its HS codes are concorded to 1996 version using the HS concordance tables.

2.2 Imports of Automation Machines and the Role of Regulating and Control Instruments

Using the Chinese Customs Data from 2000 to 2015, we first rely on the classification developed by Acemoglu and Restrepo (2022) to identify firm-level imports of machines by manufacturers. This classification identifies "machines" by HS product codes and divides machines into three types: automation machines, non-automated capital goods, and non-industrial machines. Specifically, we identify these three types of machines for HS 1996, HS 2002, HS 2007, HS 2012 version codes respectively.[2]

The Acemoglu and Restrepo (2022) classification further categorizes the three types of machines into more detailed subgroups. Among the three types, "automation machines (*Auto*)" is of the central interest in this paper, which includes industrial robots, numerically controlled machines, automatic machine tools, automatic welding machines, weaving and knitting machines, other textile dedicated machinery, automatic conveyors, and regulating and control instruments. "Non-automated capital goods (*Non auto capital*)" is capital and equipment used for similar industrial tasks but not directly related to automation, and includes "machines (*Non auto machine*)" and "tools (*Non auto tool*)", where "machines" includes heavy capital goods (including furnaces, ovens, and electrical motors), food manufacturing (machines used for brewing and baking in industrial contexts), not-numerically controlled machines, and manual welding machines, and "tools" includes tools for industrial work, manual machine tools, and tools for transferring materials. "Non-industrial machines" is machines not directly related to industrial production purposes and includes laundry machines, vending machines, agricultural machinery, and computers. Detailed descriptions of the classification can be found in Appendix A.1.

We begin our descriptive analysis by reporting the shares of three types of machines in China's total import value in Table 1. While China's total nominal import value (in US dollar) has grown by about 10 times from 2000 to 2015, imported automation machines has steadily accounted for about 3% of the total import value over the sample period. As a result, the growth in the value of imported automation machines matched the growth of China's total imports during the sample period. In contrast, the value shares accounted

[2] We thank Pascual Restrepos for generously providing the codes for identifying these goods in Acemoglu and Restrepo (2022).

for by imported non-automated capital goods and imported non-industrial machines have declined from 10.1% and 2.7% in 2001 to 6.7% and 1.0% in 2015, respectively. Figure 1 further illustrates the shares of imported automated machines and non-automated capital goods in total imports of machines for industrial production (automated machines plus not-automated capital goods) from 2000 to 2015. The share of automated machines has increased from 22.1% in 2000 to 32.2% in 2015, while the share of non-automated capital goods has shrunk from 77.9% to 67.8%. Therefore, automation machines have become more and more important in the total imports of machines and equipment.

[Table 1 & Figure 1 here]

To further investigate the roles of different subgroups in driving the growth of imported automation machines, we report the value shares of various subgroups in automation machines from 2000 to 2015 in Table 2. The striking pattern is the increasingly dominant role of "regulating and control instruments". This subgroup mostly includes measuring, controlling and monitoring instruments and appliances, and constituted 46.4% and 65.0% of China's import value of automation machines in 2001 and 2015, respectively. In terms of nominal import value, Figure 2 shows that the imported regulating and control instruments increased more than 16 times during the sample period. Further calculation based on BACI data reveals that the lion share accounted for by regulating and control instruments in automation machines is not a China-specific observation: among the world imports of automation machines in 2000 and 2015, 53.7% and 62.4% of the values are contributed by regulating and control instruments, respectively. [3] To our knowledge, the important role of regulating and control instruments among automation machines has so far been unnoticed and undocumented in the existing literature. The descriptive analysis above thus provides new evidence in this regard.

[Table 2 & Figure 2 here]

We also examine the role of industrial robots within automation machines, a subgroup of automation machines that has received disproportionately more attention of numerous

[3] When excluding China as an importer, the share of regulating and control instruments in world imports of automation machines was 54.4% in 2000 and 63.6% in 2015.

empirical studies in the recent literature. Table 2 shows that industrial robots constituted 0.6% of China's import value of automation machines in 2000, and has increased its share to 2.3% in 2015. While the import of industrial robots has achieved remarkable growth, it constitutes a small share of the total import of automation machines during the sample period.

The descriptive analysis above indicates the growing importance of automation machines and continuous automation process in China's manufacturing sector. Moreover, we uncover the dominant role of regulating and control instruments in driving the increasing imports of automation machines in China. Since regulating and control instruments are used for measuring, checking, analyzing, regulating, and controlling various physical or chemical properties, they are critical production inputs for industries that rely on precise control and monitoring, an essential element to enhance quality production capacities of manufacturing firms. Our subsequent empirical analysis thus focuses on the linkage between a manufacturer's incentive to upgrade product quality and its importation/adoption of regulating and control instruments, which has rarely been discussed by the previous literature.

3 Model: Demands for Quality and Automation

We proceed to develop a partial-equilibrium model to illustrate the impacts of demand shocks across different export markets on a firm's automation decision. With the assumptions of consumer's non-homothetic preference and the comparative advantage at quality production associated with automation technologies, the model suggests that positive demand shocks at high-income destinations induce stronger incentive to automate. We derive the main proposition to guide our empirical specifications.

3.1 A Non-homothetic Demand for Quality

We start our theoretical discussion from the demand side of the model. A firm f produces distinct varieties to serve multiple markets. We assume that the consumers' overall demand in a particular market m for the goods produced by firm f takes a standard Logit formulation:

$$q_m(f) = \exp\left[\theta_m \cdot z_m(f) - \alpha \cdot p_m(f)\right] \times D_m, \tag{1}$$

where subscript m indicates different markets with varying demands and per-capita income levels. The overall demand in market m for the variety produced and served by firm f is $q_m(f)$, with $p_m(f)$ and $z_m(f)$ being the price and quality of the variety served by firm f to consumers in m, respectively. The parameter θ_m governs the marginal utility of product quality and varies across markets m. The parameter α is the price elasticity and is assumed to be constant across different markets. The aggregate demand shifter D_m captures the macro-level demand conditions in market m, such as consumer taste, number of customers, competition environment, etc.

We assume that the demand for product quality is non-homothetic. Specifically, we assume that θ_m is monotonically increasing in the per-capita income of market m. Therefore, ceteris paribus, richer markets generate higher relative demands for high-quality goods.[4] One finds similar theoretical setups in the existing literature studying product quality in the context of international trade (Hallak, 2006; Fajgelbaum et al., 2011; Hallak and Schott, 2011; Feenstra and Romalis, 2014), and there is extensive empirical evidence in support of non-homotheticity in the preference for quality as well.

3.2 The Comparative Advantage of Automation in Quality Production

We next describe the production technology of firm f, which is affected by the quality level of goods it supplies to market m, $z_m(f)$, and its level of automation, $a(f)$. We assume the following quantity production function:

$$q_m(f) = \Phi(a(f), z_m(f)) \times l_m(f), \tag{2}$$

where $q_m(f)$ is the quantity produced by firm f for market m, and $l_m(f)$ is the amount of an elastically-supplied homogeneous input used with a fixed factor price w. The quantity production function is thus of constant returns to scale. The composite productivity term

[4] Assume that there are two varieties 1 and 2 with $\{p_1, z_1\}$ and $\{p_2, z_2\}$ and $z_1 > z_2$. In market m, the demand for variety 1 relative to that for variety 2 is:

$$q_m(1)/q_m(2) = \exp\left[\theta_m(z_1 - z_2) - \alpha(p_1 - p_2)\right],$$

which is increasing in θ_m and hence the per-capita income in m.

$\Phi(a(f), z_m(f))$ takes the following form:

$$\Phi(a(f), z_m(f)) = \phi(f) \times \Big[\frac{\exp\left[a(f) - v \cdot z_m(f)\right]}{1 + \exp\left[a(f) - v \cdot z_m(f)\right]}\Big], \quad v > 0. \tag{3}$$

It depends on firm f's quality level for market m, $z_m(f)$, its automation level, $a(f)$, and its exogenous productivity, $\phi(f)$. The firm-specific productivity component $\phi(f)$ captures the firm-level heterogeneity in production efficiency independent of quality and automation decisions.

For exposition purposes, we define $\tilde{\Phi}(a(f), z_m(f)) \equiv \Phi(a(f), z_m(f))/\phi(f)$. There are several important features of $\tilde{\Phi}(a(f), z_m(f))$ worth elaborating:

1. It is increasing in $a(f)$:

$$\frac{\partial \tilde{\Phi}(a(f), z_m(f))}{\partial a(f)} = \frac{\tilde{\Phi}(a(f), z_m(f))}{1 + \exp\left[a(f) - v \cdot z_m(f)\right]} > 0.$$

   So an increase in firm f's automation level lowers its unit production cost, given its quality decision $z_m(f)$.

2. It is decreasing in z:

$$\frac{\partial \tilde{\Phi}(a(f), z_m(f))}{\partial z_m(f)} = -\frac{v \cdot \tilde{\Phi}(a(f), z_m(f))}{1 + \exp\left[a(f) - v \cdot z_m(f)\right]} < 0.$$

   Given the technology of the firm, increasing output quality always entails a higher unit production cost.

3. It is log-supermodular:

$$\frac{\partial^2 \ln\left[\tilde{\Phi}(a(f), z_m(f))\right]}{\partial z_m(f) \partial a(f)} = \frac{v \cdot \tilde{\Phi}(a(f), z_m(f))}{1 + \exp\left[a(f) - v \cdot z_m(f)\right]} > 0.$$

   An increase in automation intensity $a(f)$ raises $\tilde{\Phi}(a(f), z_m(f))$ by a larger amount when the quality level chosen by the firm $z_m(f)$ is higher. Intuitively, this property suggests that a rise in the automation intensity generates a comparative advantage in producing high-quality goods for the firm.

Properties 1 and 2 reflect relatively common assumptions in the literature.Property 3 describes the key connection between automation technology and quality improvement. Since the production of high-quality goods puts stronger emphasis on precision and accuracy in the manufacturing process, Automation machines designated for measuring, checking, controlling, and monitoring physical and chemical properties, especially regulating and control instruments, are more critical for firms attempting to upgrade product quality. The functional form specified in (3) is reminiscent of that used by Fieler et al. (2018), in which high-quality inputs are assumed to be more productive for the production of high-quality outputs, and that used by Sun (2019), in which textile machines produced by OECD countries are assumed to be more productive in producing high-quality textile goods.

3.3 Pricing and Quality Decisions

Given the demands in different markets and the production technology, a firm f maximizes its total profit by choosing its prices, $p_m(f)$, its output quality levels, $z_m(f)$, and its automation level, $a(f)$. The market structure is monopolistic competition, so each firm faces a downward-sloping demand curve in each market, but perceives the market-level aggregates as given.

We first describe firm f's pricing and quality decisions given its automation level $a(f)$. Suppose that firm f can serve multiple markets $m \in \Omega(f)$.[5] Given quality $z_m(f)$, the firm chooses $p_m(f)$ to maximize its total profit:

$$\max_{p_m(f)} \sum_{m \in \Omega(f)} [p_m(f) - \tau_m - w/\Phi(a(f), z_m(f))] q_m(f),$$

where τ_m is a per-unit trade cost differing by m, and w is the factor price of the homogeneous input $l(f)$. Given quality level $z(f)$ and automation level $a(f)$, the pricing rule is:

$$p_m(f) = 1/\alpha + \tau_m + w/\Phi(a(f), z_m(f)). \tag{4}$$

So, the optimal price charged equals the marginal cost of production and trade plus a

[5] For simplicity, we assume that the set of markets that a firm enters, $\Omega(f)$, is fixed in our analysis. Relaxing this assumption and allowing firms to choose their $\Omega(f)$ do not affect our key theoretical results.

constant profit margin $1/\alpha$. The firm-level profit function becomes:

$$\max_{z_m(f)} \sum_{m \in \Omega(f)} \exp\left[\theta_m \cdot z_m(f) - 1 - \alpha\tau_m - \alpha w / \Phi(a(f), z_m(f))\right] \times D_m/\alpha.$$

Maximizing the profit function above with respect to $z_m(f)$ yields the following first-order condition:

$$\theta_m = -\alpha w \frac{\partial \Phi(a(f), z_m(f))/\partial z_m(f)}{\Phi(a(f), z_m(f))^2},$$

where the left-hand side is the marginal revenue of increasing $z_m(f)$ thanks to consumer's valuation to quality, and the right-hand side is the marginal cost of increasing $z_m(f)$ because quality upgrading is costly. The optimal quality choice of firm f at market m, given firm-level automation level $a(f)$, is thus:

$$z_m(f) = \left[a(f) + \ln\left(\frac{\tilde{\theta}_m \phi(f)}{\alpha w}\right)\right]/v, \tag{5}$$

where $\tilde{\theta}_m = \theta_m/v$. Equation (5) has a few implications. First, a higher level of firm-level automation induces quality upgrading. Second, a firm chooses a higher level of quality for a richer market, as $z_m(f)$ is increasing in $\tilde{\theta}_m$. Third, a higher level of firm-specific productivity $\phi(f)$ and a higher level of profit margin $1/\alpha$ both induce the firm to choose a higher quality level.

Plugging the quality decision (5) into the profit function yields the following firm-level profit function, which depends on firm f's automation level $a(f)$:

$$\pi(a(f)) = \sum_{m \in \Omega(f)} \exp\left[\tilde{\theta}_m\left(a(f) - 1 + \ln\left(\frac{\tilde{\theta}_m \phi(f)}{\alpha w}\right)\right) - \frac{\alpha w}{\phi(f)}\right] \frac{\tilde{D}_m}{\alpha}, \tag{6}$$

where $\tilde{D}_m = D_m / \exp(1 + \alpha\tau_m)$ is an adjusted aggregate demand of market m incorporating the effects of trade cost τ_m.

3.4 The Automation Decision

The automation decision of the firm f hinges on the benefit and cost associated with increasing $a(f)$. Inspecting Equation (6), we notice that the marginal benefit of increasing

the automation level $a(f)$ is:

$$\frac{\partial \pi(f)}{\partial a(f)} = \sum_{m \in \Omega(f)} s_m(f) \times \tilde{\theta}_m, \tag{7}$$

where $s_m(f) = q_m(f)/(\sum_{m'} q_{m'})$ is the share of sales generated by market m in firm f's total sales portfolio, and naturally reflects the importance of market m for firm f. Intuitively, given the same importance weight $s_m(f)$, the marginal benefit of increasing automation is larger for a richer market with a higher $\tilde{\theta}_m$, because richer consumers value high-quality goods relatively more. A larger exposure to high-income destinations therefore raises the incentive to automate.

To pin down the optimal level of automation $a(f)$, we assume that there is a convex fixed cost of automation, $C(a(f))$:

$$C(a(f)) = a(f)^{\beta}, \ \ \beta > 1. \tag{8}$$

The parameter $\beta > 1$ governs the convexity of $C(a(f))$ and describes the increasingly expensive cost of automation, such as the costs of purchasing, importing and maintaining automation machines. The fixed costs associated with automation can be justified by the empirical fact that automation or robot adopters tend to be larger than non-adopters. (Acemoglu et al., 2020; Koch et al., 2021; Ing and Zhang, 2022). More broadly, this fixed-cost formulation implies that an increase in market size/demand induces automation, consistent with the positive "market size effect" of technology adoption highlighted in the literature, e.g., Lileeva and Trefler (2010) and Bustos (2011).

The optimal automation decision thus equates the marginal benefit and the marginal cost of raising $a(f)$, and is implicitly defined by:

$$\frac{\partial \pi(f)}{\partial a(f)} = \frac{\partial C(a(f))}{a(f)} \Rightarrow \sum_{m \in \Omega(f)} s_m(f) \times \tilde{\theta}_m = \beta \cdot a(f)^{\beta - 1}. \tag{9}$$

The convexity assumption $\beta > 1$ ensures that as the weighted-average consumer valuation of quality faced by firm f, $\sum_m s_m(f) \times \tilde{\theta}_{m \in \Omega(f)}$, increases, firm f always chooses a higher automation level.

3.5 Demand Shocks and Automation

After describing the determination of automation level $a(f)$, we can now analyze the effects of demand shocks on the firm-level automation decision. We adopt a partial-equilibrium point of view and assume that changes in the aggregate market-specific demand $\tilde{D}_m$ are exogenous to the firm. Our main theoretical result is presented in the following proposition:

Proposition 1. The elasticity of firm f's automation level with respect to the demand shock at market m is:

$$\frac{\partial \ln a(f)}{\partial \ln \tilde{D}_m} = s_m(f) \times \frac{\tilde{\theta}_m}{\Theta(f)}, \tag{10}$$

where $\Theta(f) = (\beta - 1)[\sum_{m \in \Omega(f)} s_m(f) \cdot \tilde{\theta}_m] - [\sum_{m \in \Omega(f)} s_m(f) \cdot \tilde{\theta}_m^2]a(f)$ is a firm-specific parameter and $\Theta(f) > 0$ according to firm f's second-order condition in determining $a(f)$.

Equation (10) shows that given the sales share of market m for firm f, $s_m(f)$, a positive demand shock in market m generates a larger effect on firm-level automation when consumers in market m have a higher marginal valuation of product quality with a higher value of $\tilde{\theta}_m$. Put it another way, since $\tilde{\theta}_m$ is increasing in the per-capita income of market-m consumers, given the sales share $s_m(f)$, rising product demands in high-income destinations stimulate stronger increases in automation by producers.

To summarize, our model offers novel testable predictions about how the impacts of demand shocks in different markets on firm-level automation decisions vary systematically with the per-capita income levels of these markets. Several existing studies have indeed found that demand shocks in high-income export markets also cause larger increases in skilled workers hiring (Brambilla et al., 2012) and intermediate input prices (Bastos et al., 2018) with similar theoretical interpretations through the lens of non-homotheticity and product quality, but our focus on the impacts on firm-level automation as a technology adoption decision is distinct. The subsequent empirical analysis will be devoted to taking the key insights of equation (10) to empirical examination.

4 Empirical Strategy

The main theoretical prediction that we take to the data is the income-related heterogeneity in elasticity described by (10) in Proposition 1. For empirical implementation, we group

all export markets, based on their per-capita income levels, into a high-income set, $\Omega(f)_H$, and a low-income set, $\Omega(f)_L$, such that $\Omega(f)_H \cup \Omega(f)_L = \Omega(f)$ and $\Omega(f)_H \cap \Omega(f)_L = \varnothing$. Furthermore, we impose an assumption that

$$\begin{aligned} \tilde{\theta}_m &= \tilde{\theta}_H, \quad \forall m \in \Omega(f)_H, \\ \tilde{\theta}_m &= \tilde{\theta}_L, \quad \forall m \in \Omega(f)_L. \end{aligned} \tag{11}$$

So we basically simplify the model into a two-market setup: a high-income destination H and a low-income destination L. Our theory thus predicts that, the elasticity of firm f's automation with respect to demand shock in market m adjusted by firm f's sales share in market m is higher for $m = H$:

$$\frac{1}{s_H(f)} \cdot \frac{\partial \ln a(f)}{\partial \ln \tilde{D}_H} = \frac{\tilde{\theta}_H}{\Theta(f)} > \frac{\tilde{\theta}_L}{\Theta(f)} = \frac{1}{s_L(f)} \cdot \frac{\partial \ln a(f)}{\partial \ln \tilde{D}_L}. \tag{12}$$

We use the implication of equation (12) in the simplified two-market setup to guide our empirical strategy, namely, the adjusted elasticity for high-income destination is larger than that for low-income destination.

4.1 Baseline Specification and Construction of Key Variables

Our objective is to examine how demand shocks in high- and low-income destinations affect firm-level automation adoption. Guided by the theoretical prediction in equation (12), we estimate a first-difference specification that removes time-invariant firm characteristics, similar to Aghion et al. (2024) and Demir et al. (2024):

$$\Delta \ln y_{ft} = \beta_H \cdot (s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft}) + \beta_L \cdot (s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft}) + \mathbf{b} \cdot \mathbf{X}_{ft} + \epsilon_{ft}. \tag{13}$$

Our main outcome variable $\Delta \ln y_{ft}$ is the log change of firm-level import of automation machines, as identified in Section 2.2 following the approach of Acemoglu and Restrepo (2022). Admittedly, data limitations prevent us from identifying automation machine purchases from domestic suppliers within China. However, given that quality-enhancing machines primarily originate from high and upper-middle income economies (rather than from Chinese domestic producers) during our sample period, we expect that imports represent the

majority of the purchases and installations of automation machines.[6]

The log-change specification requires that firms report imports in consecutive years.[7] To account for the potential role of the extensive margin, we also consider the change in import status in our baseline result and a mid-point growth approach in robustness analyses to incorporate entry and exit.

The high-income group, denoted by subscript H, includes countries identified as high- or upper-middle-income in the 2001 World Bank income classification, while the low-income group, denoted by subscript L, includes countries classified as low- or lower-middle-income in the same classification. The sets Ω_H and Ω_L correspond to these high- and low-income destination countries respectively.[8] The export demand shocks in high- and low-income markets faced by firm f, adjusted by its sales shares are defined as follows:[9]

$$\begin{aligned} s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft} &= \sum_h \sum_{c \in \Omega_H} (s_{fch,t-1} \times \Delta \ln M_{cht}), \\ s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft} &= \sum_h \sum_{c \in \Omega_L} (s_{fch,t-1} \times \Delta \ln M_{cht}). \end{aligned} \tag{14}$$

For high-income and low-income destinations, the adjusted export demand shock is the weighted average demand shock across all product-destination pair in high-income and low-income economies, respectively.[10] We define a product h as a unique HS 4-digit code.[11] The log change $\Delta \ln M_{cht}$ is the log change of total manufacturing imports of product h by country c from all countries other than China, from year $t-1$ to year t. The data to construct $\Delta \ln M_{cht}$ comes from the BACI Dataset (202401b Version). The weight $s_{fch,t-1}$

[6] For example, Acemoglu et al. (2020), Ing and Zhang (2022) and Bonfiglioli et al. (2024) all rely on imports of robots or automation machines to capture firm-level automation decisions.

[7] In our sample, about half of the machine-importing firms also imported regulating and control instruments for two consecutive years.

[8] We use 2001 World Bank income classification to reduce concerns of endogeneity. We drop countries or regions not in BACI or do not have income classification in the World Bank. These observations account for less than 2% of import and export value. The countries or regions dropped are in Appendix A.2.

[9] Demir et al. (2024) has a similar construction but use GDP per capita to interact with demand shocks.

[10] Due to the lack of information on domestic sales at the firm-product level, we focus on the high-income and low-income export markets in our baseline analysis. Therefore, the shares on the left-hand side of (14), $s_{H,f}$ and $s_{L,f}$, are the sales share of high-income and low-income destinations for firm f's total export, so we have $s_{H,f} = \sum_h \sum_{c \in \Omega_H} s_{fch,t-1}$ and $s_{L,f} = \sum_h \sum_{c \in \Omega_L} s_{fch,t-1}$, and hence $s_{H,f} + s_{L,f} = 1$. In one of our robustness analysis, we also try to account for the potential impacts of domestic demand shock in China.

[11] As in Demir et al. (2024), we aggregate trade flows to HS-4 digit level because trade flows exhibit higher volatility in HS-6 digit level data.

denotes the share of firm f's export of product h to country c in firm f's total export sales value across all product-country pairs in year $t-1$.

The key focus of our empirical analysis is to test the theoretical prediction of equation (12): $\frac{\tilde{\theta}_H}{\Theta(f)} > \frac{\tilde{\theta}_L}{\Theta(f)}$. The empirical counterpart of this prediction is whether $\beta_H > \beta_L$, which requires consistently estimating β_H and β_L in equation (13). To achieve this, we also include a set of control variables $\mathbf{X}_{ft}$ in equation (13) to account for any potential confounding factors that may introduce bias in our estimates of β_H and β_L. We postpone the discussion of the related details to Subsection 4.2.

The regression sample for our empirical analysis contains only Chinese manufacturing exporters (excluding trade intermediaries and agriculture exporters). We also construct and use demand shocks for only manufacturing products.[12] Given the first-difference specification in equation (13), our baseline regression sample spans the period from 2002 to 2015, the period after China's WTO accession.

Figure 3 depicts the distribution of automation machine imports across firms, grouped by their export share to high- and upper-middle-income markets ($share_high$ in the figure). Firms exporting exclusively to these markets ($share_high = 100\%$) show a gradual rise in their automation import share—from 28% in 2001 to 32% in 2015. By 2015, roughly 65% of automation imports were undertaken by firms whose export shares to these markets ranged between 0% and 100%, representing the main source of variation in firm-level import share.

4.2 Identification and Potential Confounding Factors

It is crucial to discuss what consistent estimates of β_H and β_L in equation (13) entail. Our key explanatory variables $s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft}$ and $s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft}$ essentially fall into the category of the "shift-share design" approach. In our context, according to Borusyak et al. (2022), firms with positive or negative unobserved automation shocks ϵ_{ft} may systematically specialize in certain product-destination pairs ch, so any potential correlation between ϵ_{ft} and $s_{fch,t-1}$ does not matter for consistency. Put it differently, endogeneity of the "weights" $s_{fch,t-1}$ does not necessarily cause inconsistency.

[12] A firm's primary industry is identified using the largest HS2 sector in which the firm exports from 2001 to 2015. We identify manufacturing industries and products using SITC manufacturing classification and its concordance to HS codes.

What really matters for the consistency of $\hat{\beta}_H$ and $\hat{\beta}_L$ is that for product-destination pairs ch specialized by firms with higher or lower unobserved automation shocks (such as lower cost of importing machines, greater adaptability to automated production, or better access to automation-related technology), their demand shocks $\Delta \ln M_{cht}$ cannot be systematically higher or lower.[13] We view the demand shocks $\Delta \ln M_{cht}$ at the product-destination level as exogenous to firm-level unobserved automation shocks ϵ_{ft}, so our baseline specification (equation 13) can consistently estimate β_H and β_L.

Identification also requires controlling for other potential confounding factors. We include the following control variables into $\mathbf{X}_{ft}$:

1. Exchange rate: Varying exchange rate may affect a firm's automation behavior. On one hand, a depreciating Chinese Yuan (CNY) increases the competitiveness of Chinese exporters, which may generate a positive "market size" effect facilitating automation adoption. On the other hand, a depreciating (CNY) raises the cost of importing, discouraging the imports of automation machines. We follow Dai and Xu (2017) to construct the log changes of firm-specific import and export exchange rates, $\Delta \ln XR^M_{ft}$ and $\Delta \ln XR^X_{ft}$:

$$\Delta \ln XR^M_{ft} = \sum_c (\omega^M_{fc,t-1} \times \Delta \ln XR_{ct}),$$
$$\Delta \ln XR^X_{ft} = \sum_c (\omega^X_{fc,t-1} \times \Delta \ln XR_{ct}),$$

   where $\omega^M_{fc,t-1}$ ($\omega^X_{fc,t-1}$) is the share of firm f's value of import from (export to) country c in the firm's total import (export) value in $t-1$. The variable XR_{ct} is the nominal exchange rate of country c's home currency against CNY in year t, obtained from the IMF and expressed in indirect quote. So an increase in XR_{ct} indicates an appreciation of country c's home currency against CNY.

2. Import tariff: As perhaps the most important policy [pls check] barrier to trade, import tariff of automation machines also affects firm-level automation imports. We construct the log change of firm-specific import tariff for product group G, which may represent

[13] For example, firms intensively exporting to Japan may receive more industrial knowledge about automation due to Japan's leading role in manufacturing robots, but the growths of demands for goods in Japan cannot be systematically higher or lower than those in other high-income destinations.

automation machines, non-automated capital goods, or any subgroup of them:

$$\Delta \ln tar_{ft}^{G} = \sum_{k \in \Omega^{G}} (\omega_{fk,t-1}^{G} \times \Delta \ln (1 + tar_{kt})).$$

In our baseline regression, we examine the impacts of demand shocks on imported automation machines, so G represents automation machines (*Auto*) and its four subgroups: other automated machines (*Auto machine*) and automated tools (*Auto machtool*) and regulating and control instruments (*Regulate*).[14] The weight $\omega_{fk,t-1}^{G}$ denotes the value share of firm f's import of product k within its import of the whole product group G in year $t-1$. The variable tar_{kt} denotes the China's effectively applied tariff on product k in year t. Tariff data are from WITS TRAINS.[15]

3. Domestic demand shock: One important potential confounding factor is China's domestic demand shock, which may also drive automation given China's large domestic market. As noticed in the previous section, the lack of information on domestic sales at firm-product level prevents us from constructing reliable measures for domestic demand shock in the baseline specification. In one of our robustness analysis, we try to control for the potential effect of domestic demand shock using an imperfect yet the most granular measure available to us.

   Specifically, we complement our analysis using the Chinese Annual Survey of Industrial Firms (ASIF) data from 2000 to 2013. First, we construct domestic demand shocks at the industry (HS 2-digit)-province-year level by calculating the first-difference log change of the leave-one-province-out total domestic sales following Lim et al. (2022), to try to isolate industrial sales growth driven by national demand but not supply-side forces of the particular province that a firm locates in.[16] Second, we calculate the share of exports in a firm's total sales using the ASIF data. Export data come

[14] In one of our placebo tests, we examine the impacts of demand shocks on non-automated capital goods and non-industrial machines. Thus, we also construct firm-specific import tariffs for non-automated capital goods (*Non auto capital*) and its subgroups: non-automated machines (*Non auto machine*) and non-automated tools (*Non auto tool*), and for non-industrial machines ($Non - industrial\ machine$) and its subgroups: computer (*Computer*) and other non-industrial machines ($Other\ non - industrial\ machine$).

[15] WITS defines effectively applied tariff as the lowest available tariff. If a preferential tariff exists, it is used as the effectively applied tariff. Otherwise, the MFN applied tariff is used.

[16] We concord the industry codes used in ASIF data and HS 2-digit codes following Brandt et al. (2017).

from the Chinese Customs Data as it provides a more accurate measure for trade than the ASIF, while total sales data comes from the ASIF. Third, we construct the firm-specific domestic demand shock adjusted by firm f's sales share in the domestic market as follows:

$$s_{CHN,f,t-1} \times \Delta \ln \tilde{D}_{CHN,ft} = s_{CHN,f,t-1} \times \Delta \ln Y_{i(f),p(f),t}, \tag{15}$$

where $s_{CHN,f,t-1}$ denotes the share of domestic sales in firm f's total sales (domestic sales share) in year $t-1$. Firm-level domestic sales are calculated as firm total sales minus firm exports from the Customs Data.[17] Consequently, the domestic sales share is calculated as one minus firm f's export-to-sales ratio. If firm f's export-to-sales ratio is missing (possibly due to the incomplete or imperfect matching between the Customs data and the ASIF data), we use the corresponding industry-city-year-specific or industry-province-year-specific average export intensity to impute the share.[18] For firm f operating in industry i and locating in province p, $\Delta \ln Y_{i(f),p(f),t}$ denotes log change of the leave-one-province-out total sales of HS 2-digit industry i and province p from year $t-1$ to t, measuring the domestic demand shock faced by firm f.

4. Labor cost: Previous studies highlight the motive to save labor cost as an important driving force for automation, so we also construct and control for city-level changes in labor cost. We use the log change of wage ($\Delta \ln \text{Wage}_{ct}$) at the city-year level following Fan et al. (2021). Data come from the National Bureau of Statistics in China.[19]

In addition to the control variables that we described above, we also control for industry-province-year fixed effects in all our regressions, so we effectively control for any time-varying supply- or demand-side factors specific to an industry-province combination. We thus exploit

[17] Before calculating firm-level domestic sales, we need to undertake four steps to clean and match the Customs data and the ASIF data: First, we link firms across 2000-2013 in the ASIF data following Brandt et al. (2012). Second, we use linear imputation for missing sales in the ASIF data. Third, we filter ASIF following Feenstra et al. (2014). Fourth, we merge the Customs Data and the ASIF data using firm names and firm zip code and telephone number following Yu (2015). The merged dataset covers 174,652 firms (20.2% of ASIF firms, 26.1% of Customs Data firms) and 77.5% of export value in the filtered ASIF data. Our matching quality is comparable to Yu (2015) and details are reported in Appendix Table A1.

[18] HS 2-digit industry is denoted by the largest HS 2-digit industry that a firm exports from 2001-2015.

[19] We employ linear imputation for missing wage observations. Whenever city-year-level wages are missing after within-city linear imputation, we use province-year-level wages as proxies.

the over-time variations across firms within an industry-province pair to identify β_H and β_L. Standard errors are clustered at the firm level, except for one of the robustness analyses where we report bootstrapped standard errors.

5 Empirical Results

5.1 Baseline Results

Table 3 examines the impacts of high-income and low-income demand shocks on Chinese firms' imports of both general machines and automation machines, utilizing various constructions of the dependent variables. Columns 1 and 2 focus on the extensive margin, where the dependent variable is the change of the dummy variable indicating whether a firm imports machines or automation machines as follows

$$\Delta \mathbf{1}(y_{ft} > 0) = \beta_H \cdot (s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft}) + \beta_L \cdot (s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft}) + \mathbf{b} \cdot \mathbf{X}_{ft} + \epsilon_{ft} \quad (16)$$

where $\mathbf{1}(y_{ft} > 0)$ is the dummy variable indicating whether firm f imports a specific type of machine y.

Columns 3 and 4 estimates equation (13) and analyzes the intensive margin with the log difference of imported machines and automation machines. Columns 5 and 6 analyze the perpetual inventory measure, using the log difference of the imputed stock of imported machines or automation machines calculated following Brandt et al. (2012).

Export demand is not the only driver for automation, domestic demand from the large Chinese markets can also drive automation adoption. Thus, column 7 and 8 reports the results accounting for our constructed domestic shocks. The dependent variables are the log changes of automation machines (column 3 and 4). Domestic shocks $\Delta \ln \tilde{D}_{CHN,ft}$ are constructed in Equation (15), and interacted with firm-level domestic sales $s_{CHN,f,t-1}$. Sales data come from the Annual Survey of Industrial Firms (ASIF). Column 7 repeats the baseline regression in column 4 using the Customs-ASIF merged sample. Column 8 further controls for domestic demand shock. The results remain robust after accounting for domestic demand shocks.

The estimated $\beta_H - \beta_L$ is positive but not significant in column 1 for the extensive

margin of imported machines and in column 3 for the intensive margin of imported machines. It turned significant for column 5 when using imputed capital stock of total machines. However, in column 2, 4, 6-8, the estimated $\beta_H - \beta_L$ is statistically significant and positive for automation machines.

[Table 3 here]

Previous studies have documented a "market size" effect of demand shock on technology adoption purely driven by increases in export market size (Lileeva and Trefler, 2010; Bustos, 2011; Aghion et al., 2024). If market size is the sole driver of firm automation, rather than high-income demand shocks, we would expect both the adjusted high-income and low-income demand shocks to generate statistically significant and economically similar impacts. However, our results highlight a significant difference in the estimated β_H and β_L. The estimated coefficients for high-income destinations β_H are positive and statistically significant at the 1% level for machines and automation machines. For example, column 4 indicates that a 10% increase in demand from high-income destinations leads to a 2.1% increase in firm-level imports of automation machines. In contrast, demand shocks from low-income destinations have no statistically significant impact on the imports of automation machines. except 10% significance for column 7-8 in the Customs-ASIF merged data

Table 5 presents robustness tests for demand shocks' impact on automated machine imports. Column 1 takes the extensive margin into consideration by following Aghion et al. (2024) to construct the mid-point growth rate for both the explanatory variables and the outcome variables. The export demand shocks in market m accounting for extensive margin (mid-point growth) for firm f between t and $t-1$ is constructed as:

$$s_{m,f,t-1} \times \Delta\tilde{D}_{m,ft} = \sum_{h} \sum_{c \in \Omega_m} (s_{fch,t-1} \times \left(\frac{M_{hc,t} - M_{hc,t-1}}{\frac{1}{2}(M_{hc,t} + M_{hc,t-1})} \right)) \tag{17}$$

where $market$ $=H$ or L. The definitions of $s_{fch,t-1}$ and M_{cht} are the same as those in equation (14). The firm-level change in the imports of automation machines is also calculated using mid-point growth specification: $\frac{M_{G,ft} - M_{G,ft-1}}{\frac{1}{2}(M_{G,ft} + M_{G,ft-1})}$, where G denotes general machines ($Machine$) and automation machines ($Auto$). The number of observation in column 1 is larger than those in Table 3 because allowing for entry and exit mechanically increases

effective sample sizes. Our result is robust after accounting for the extensive margin of trade in column 1. Column 2 uses t-2 period demand shocks, Column 3 reports bootstrap standard errors. Both column 2 and 3 show significantly positive coefficient differences. Column 4 examines trading companies, yielding no significant effects as expected, while Columns 5 and 6 show no significant impacts on non-automation and non-industrial machines, confirming the specificity of results to automation machines.

[Table 5 here]

5.2 Exploring Mechanisms

5.2.1 Specific Automated Machines: Regulating and Control Instruments

Table 6 examines the effects of demand shocks on specific types of automation machinery: robots (Column 1), other automated machines (Column 2), automated machine tools (Column 3), and regulating and control instruments (Column 4). The coefficients for both demand shocks are statistically insignificant for robots and other automated machines. For automated machine tools, the coefficient on high-income market demand shocks is positive and significant, although the estimated difference $\beta_H - \beta_L$ is positive but insignificant. Only regulating and control instruments exhibit a significantly positive $\beta_H - \beta_L$, suggesting that this category primarily drives the overall impact of demand shocks on automation-related imports.

[Table 6 here]

Appendix Table B1 reports robustness checks for regulating and control instruments across alternative specifications: (i) using the merged Customs–ASIF dataset (Column 1), (ii) controlling for domestic demand (Column 2), (iii) employing demand shocks lagged by two periods (Column 3), and (iv) applying bootstrapped standard errors (Column 4). The results remain robust across all specifications.

Appendix Table B2 presents placebo tests for imported non-automated capital goods and non-industrial machinery.[20] We find that while positive demand shocks increase firm-level imports of non-automated capital goods, the coefficient difference $\beta_H - \beta_L$ is never

[20] The dependent variables in Columns 1–3 are the log changes in imported non-automated capital goods (Column 1), non-automated machines (Column 2), and non-automated tools (Column 3). Columns 4–6

statistically significant. Moreover, imports of non-industrial machinery show no response to demand shocks. These findings indicate that the baseline effects are specific to automation machinery—particularly regulating and control instruments—and do not extend to non-automation equipment.

As an additional placebo, Appendix Table B3 replicates the analysis for the log changes in imported automation machinery and its subcategories among trade intermediaries, while Table B4 repeats the placebo tests for non-automated and non-industrial machinery.[21] Since intermediaries do not produce export goods, demand shocks from high-income destinations should not affect their imports of production machinery. Consistent with this prediction, both the demand shock coefficients and the $\beta_H - \beta_L$ differences are statistically insignificant for intermediaries, regardless of machinery type. Likewise, demand shocks do not significantly influence imports of non-automated or non-industrial equipment among trading firms.

Finally, Appendix Table B5 presents results using firms' export shares in $t-2$ to construct sales shares in the shift-share design. Table B6 reports the baseline results using bootstrapped standard errors. Across all robustness checks, our main findings remain robust.

5.2.2 Quality Upgrading

A key mechanism in our theoretical analysis is firm's motivation of quality upgrading, so we follow up by directly investigating the impacts of demand shocks on firm export prices and quality. We construct firm-product(HS 6-digit)-year-level export price as simple average of firm-product(HS 6-digit)-destination-year export prices. Prices are trimmed at both the upper and lower 1% levels for each HS 6-digit product-year group. We also infer export quality following Khandelwal et al. (2013) (KSW hereafter), who derive a specification for estimating quality in a CES (Constant Elasticity of Substitution) preference. Firm f exports HS6 product h to destination d in year t. Let σ indicate the elasticity of substitution, q and

use the log changes in imported non-industrial machinery (Column 4), computers (Column 5), and other non-industrial machines (Column 6) as dependent variables. Detailed variable definitions are provided in Appendix A.1.

[21] Trade intermediaries are identified following Ahn et al. (2011), Poncet and Xu (2018), and Tang and Zhang (2012). Specifically, we search firm names for pinyin (Romanized Chinese) terms meaning "trading," "importer," or "exporter," including "jin4chu1-kou3," "jing1mao4," "mao4yi4," "ke1mao4," "wai4jing1," "wai4mao4," and "gong1mao4."

p indicate quantity and price respectively. The specification is:

$$\ln q_{fhdt} + \sigma \ln p_{fhct} = \alpha_{hdt} + \epsilon_{fhdt} \tag{18}$$

where α_{hdt} is a product-country-year fixed effect.[22] Quality is estimated within each sector following Lim et al. (2022). The measured quality within a product-destination-year triplet is $\lambda^*_{fhdt} = \frac{\epsilon_{fhdt}}{\sigma - 1}$. The intuition is that quality is an attribute that can increase prices without lowering market shares. The estimates for elasticity σ come from the median world elasticity for each HS 3 digit industry from Broda et al. (2006). We aggregate quality at firm-product-country-year ($fhct$) level to firm-product-year (fht) level following Lim et al. (2022). Table 9 shows that demand shocks from high-income destinations raise export prices and export quality more than demand shocks from low-income destinations, with the difference being statistically significant at the 5% level for export price. Therefore, we observe consistent evidence of quality upgrading when firms increase their imports of automation machines as reaction to expanding demands in high-income destinations.

[Table 9 here]

5.3 Heterogeneity Analysis

5.3.1 Sector Characteristics

This section examines whether the impact of demand shocks on firms' automation imports varies across industries. We classify industries at the HS 2-digit level, which provides a meaningful balance between sectoral detail and comparability. Each firm is assigned to the HS2 industry in which it records the largest export share during 2001–2015. Table 7 analyzes heterogeneity across HS2 sectors categorized by Rauch (2001) and price dispersion. Following Cadot et al. (2013), price dispersion is measured by the coefficient of variation of prices at HS-2 digit level. Firms are divided into three groups, namely differentiated goods, reference goods and homogenous goods sector, according to Rauch (2001), and further classified into two groups based on the median of price dispersion across industries: high dispersion, indicating substantial price and quality differences, and low dispersion, reflecting

[22] KSW use product and country-year fixed effects. Here we posit that quality may have common components within each product-country-year market, justifying the inclusion of a more granular fixed effect.

relatively uniform price and quality. We re-estimated equation (13) for the 5 sub-samples described above, and report the test of coefficient difference $\beta_H - \beta_L$ for imported regulating & control instruments. Figure 4 presents the results, with 90% (as dots) and 95% (as bars) confidence intervals. Each point represents the estimated $\beta_H - \beta_L$. Coefficient differences are significant for differentiated and reference goods sectors, as well as high-price-dispersion sectors, indicating stronger effects in differentiated and high-dispersion sectors. In these markets, high-income markets reward higher quality, consistency, and compliance with stricter standards, which raises the marginal return to automation.

[Figure 4 here]

Impacts of demand shocks across HS2 sectors are reported in Appendix Table B7. Figure 8 illustrates the results of the coefficient difference test for each regression, displaying the 90% and 95% confidence intervals for imported regulating and control instruments across various aggregated industries. These industries are defined according to HS 2 sectors: the Agricultural sector (HS codes 1–24) and the Mining sector (HS 25–27) were excluded due to insufficient observations. Other sectors include Chemicals (HS 28–38); Plastics and Other Materials (HS 39–49); Textile and Footwear (HS 50–67); Cement & Metal Products (HS 68–83); Machines (HS 84); Electronics (HS 95); Vehicles (HS 86–89); Optical Instruments (HS 90–92); and Furniture, Toys, and Others (HS 93–98). The results show that significant coefficient differences are found in Chemicals, Cement & Metal Products, and Electronics, consistent with results for differentiated and reference goods sectors. Firms in Chemicals, Cement and Metal products, and Electronics respond more strongly to high-income demand because these industries are highly quality-sensitive, requiring strict precision, reliability, and consistency that automation can deliver. Buyers in the Chemicals and Cement & Metal Products industries have high demands for quality consistency (e.g., cement and metal products are used in infrastructure, automotive, and aerospace applications). Moreover, in the Electronics and Chemicals sectors, automation can significantly reduce human errors and improve the precision of complex processes (e.g., electronic assembly and chemical formulations).

[Figure 8 here]

5.3.2 Firm Characteristics

Table 8 further explores heterogeneity by firm size and ownership. For firm size, we classify firms into two groups—those with initial export values above the median (large firms) and those below the median (small firms). For ownership, firms are categorized into three types: state-owned enterprises (SOEs), foreign-invested enterprises (FIEs), and private and other firms (e.g., collective firms and other types).[23]

Figure 6 displays the results of coefficient difference tests for the impact of demand shocks on imported regulating and control instruments, presented with 90% and 95% confidence intervals. Larger firms and foreign-invested enterprises (FIEs) show significantly positive coefficient differences, while state-owned enterprises (SOEs) and private firms do not, highlighting FIEs' greater responsiveness to demand shocks.

[Figure 6 here]

We now examine the dynamic effect of the demand shock using the following equation. Notations of outcome variables and control variables are the same as equation (13).

$$\Delta \ln y_{f,t} = \left(\sum_{\tau=-k'}^{k} \alpha_\tau \Delta D_{H,f,t-\tau} \right) + \left(\sum_{\tau=-k'}^{k} \alpha_\tau \Delta D_{L,f,t-\tau} \right) + \mathbf{b} \cdot \mathbf{X}_{ft} + \epsilon_{ft} \tag{19}$$

where $\Delta D_{M,f,t-\tau} = s_{M,f,t-\tau-1} \times \Delta \ln \tilde{D}_{M,ft\tau}$ and $M = H, L$, k=2.

Figure 10 presents the coefficients of high- and low- demand shocks and the results of the coefficient difference test for equation (19), examining the dynamic effects of the demand shocks on imported automation machines and regulating & control instruments respectively, along with the 90% (as dots) and 95% (as bars) confidence intervals for imported automated machines. For automation machines, only the contemporaneous effects of both high- and low-income demand shocks are positive and significant, while no dynamic effects are observed. For regulating & control instruments, only the contemporaneous effect of high-income demand shocks is significant, whereas low-income shocks show no significant effect. Regarding the estimated $\beta_H - \beta_L$, the contemporaneous difference is positive but not significant for automation imports, yet significantly positive for regulating & control

[23] We identify firm ownership based on the sixth digit of their registration code in the Customs Data, which shows the ownership type upon registration.

instruments even when the sample size is reduced by half after including two leads and two lags of demand shocks.

6 Conclusions

This paper investigates how export market demands drive automation adoption in China's manufacturing sector. We argue that increasing demand from high-income consumers, who prioritize quality, incentivizes producers to automate more actively. Our theoretical model generates this insight, suggesting that automation becomes more attractive as demands for higher-quality goods rise.

Using data on Chinese firm exports and imports from 2001 to 2015, we empirically test our theoretical prediction. We find that positive demand shocks from high-income destinations, but not low-income destinations, significantly increase a firm's imports of automation machines, particularly regulating and control instruments, which are crucial for enhancing production precision and reliability. Furthermore, the responses in imported automation machines are primarily driven by imports sourced from high-income economies. Our finding indicates that the motive to satisfy high-income consumers' need for high-quality goods is a critical force to promote technology adoption, and this force should be taken into consideration in the making of trade policies, especially for developing economies.

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Figures

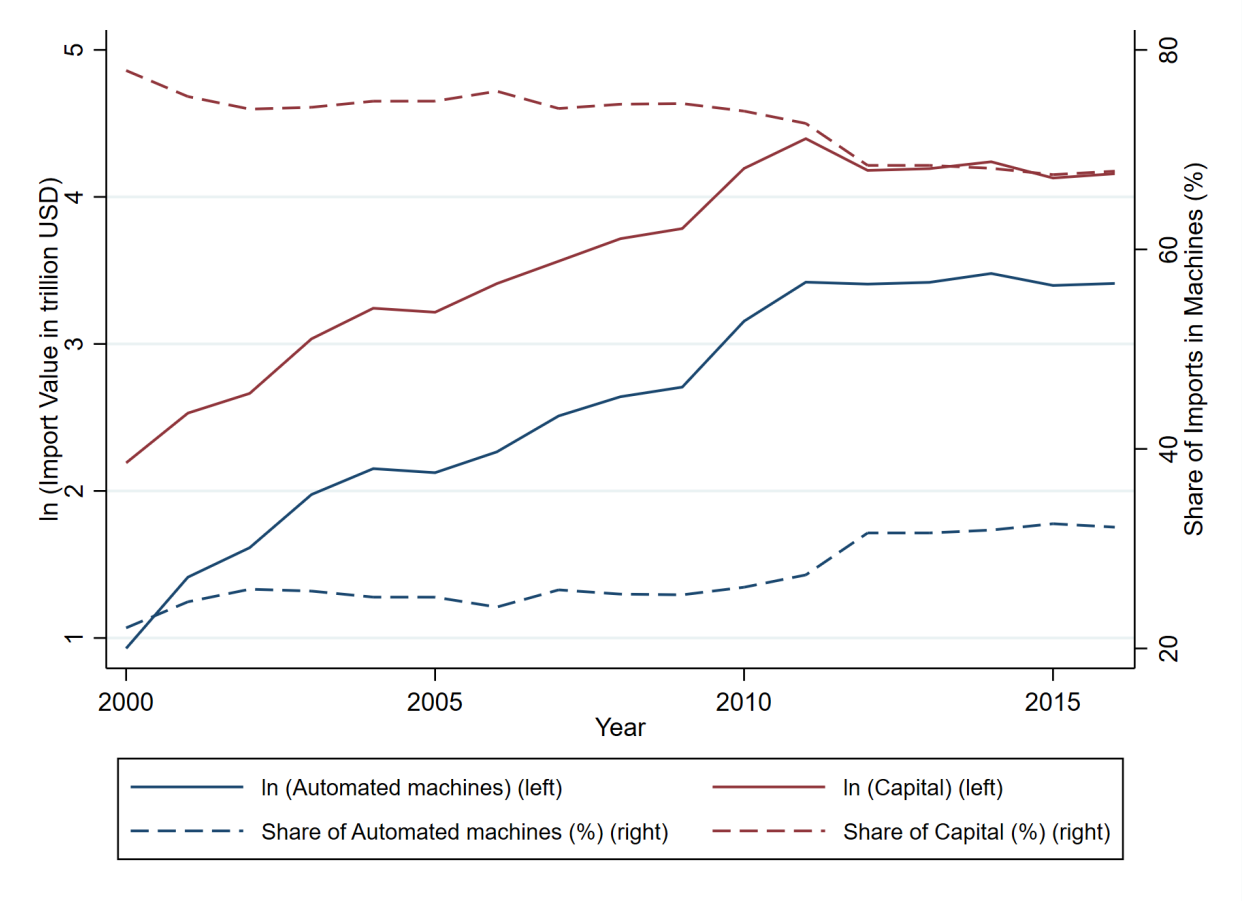


Figure 1: Imports of Automated Machines and Non-automated Capital Goods in China (2000-2015)

Notes: This figure plots the logged import values of automated machines and non-automated capital goods (labeled as "Capital" in the figure) on the left axis, and their respective shares in total machine imports (automated plus non-automated capital goods) on the right axis, for the period 2000–2015. The share of automated machines increased from 22.1% in 2000 to 32.2% in 2015, while the share of non-automated capital goods declined from 77.9% to 67.8%. All statistics are calculated by the authors based on Chinese Customs data.

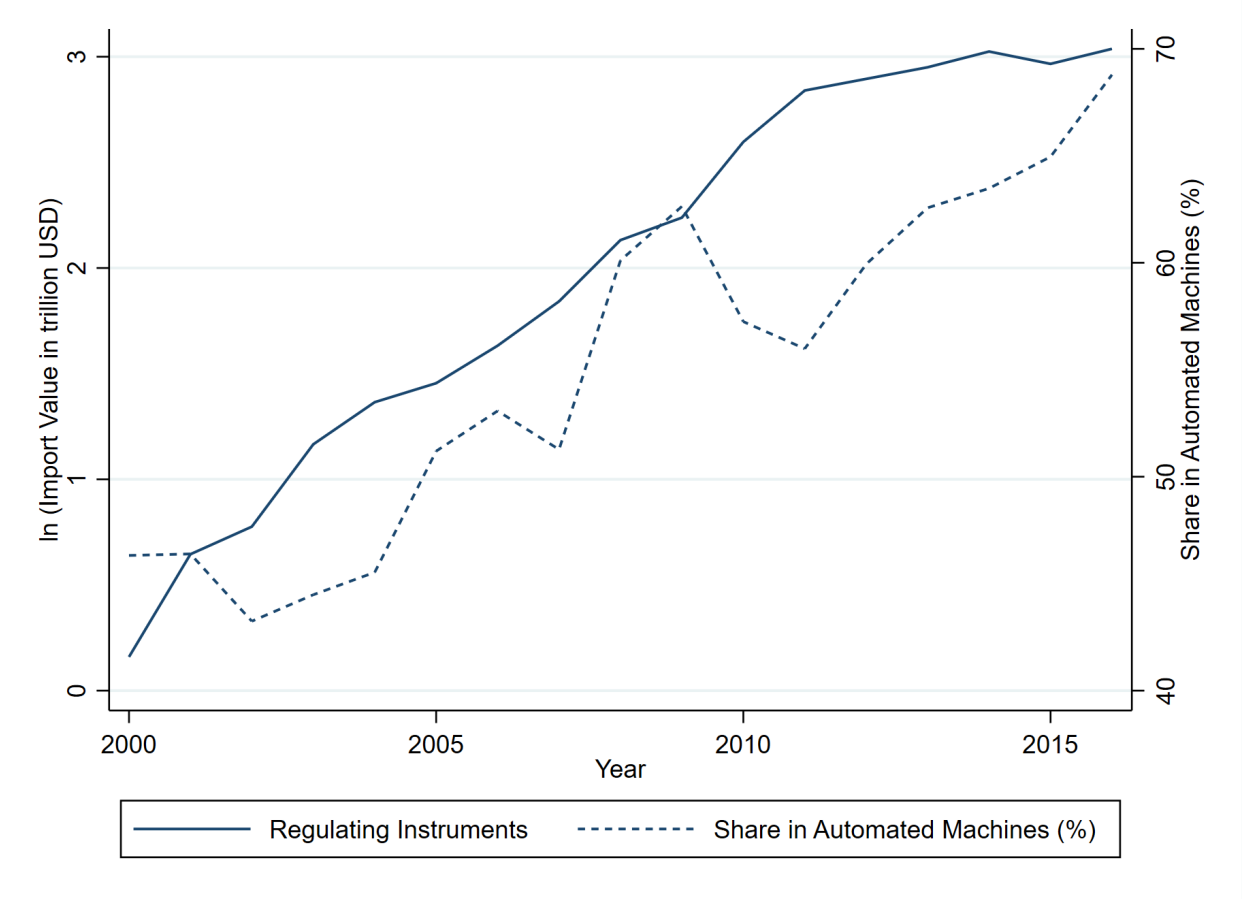


Figure 2: Imports of Regulating and Control Instruments in China (2001-2015, Regression Sample)

Notes: This figure reports the logged value of imported regulating and control instruments and its share in automated machines. The share of regulating and control instruments increased from 46.3% in 2000 to 65.0% in 2015. This highlights the importance of regulating and control instruments in imported machines, which is rarely discussed in previous literature. The statistics are calculated by authors from the Chinese Customs Data.

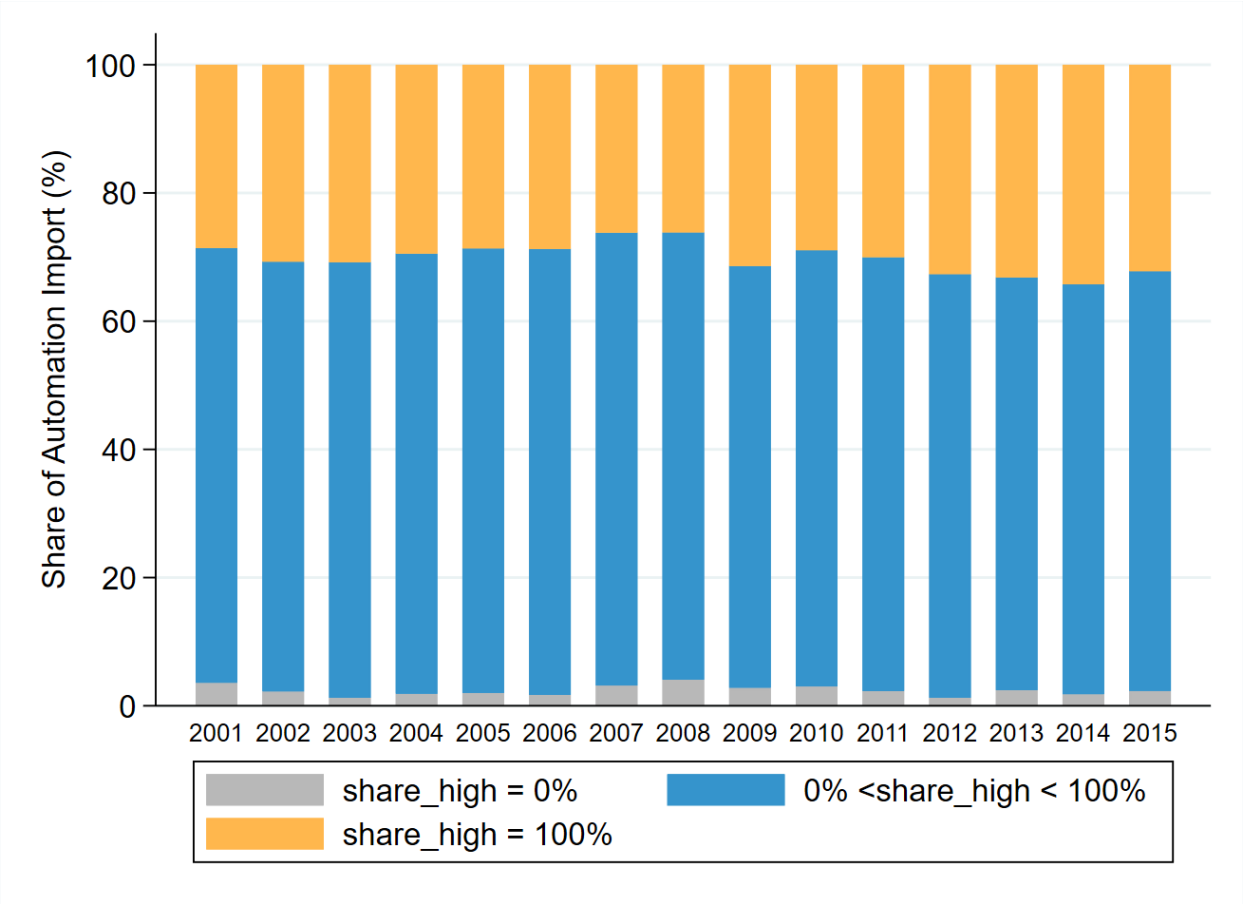


Figure 3: Imports of Automation Imports and Firms' Share of Exports to High and Upper-Middle Income Markets (2001-2015)

Notes: This figure illustrates the share of automation machine imports across firms, categorized by their share of exports to high- and upper-middle-income markets (share_high in the figure). Firms exporting exclusively to these markets (100% share) demonstrate a slow increase in automation machine import share over time from 28% in 2021 to 32% in 2025. In 2015, approximately 65% of automation imports were conducted by firms with export shares ranging from 0% to 100% to these markets. The sample is restricted to ordinary trade and excludes non-trading companies.

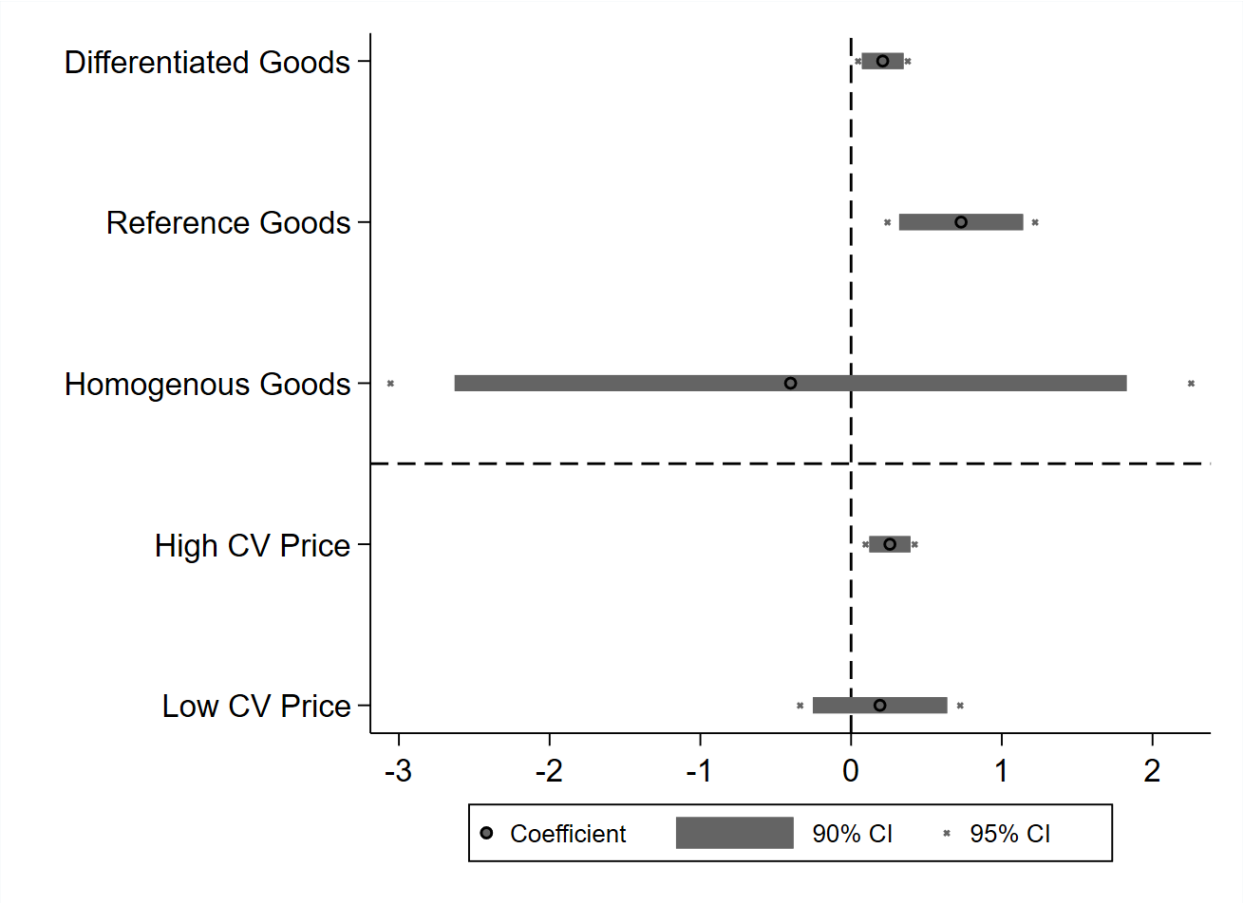


Figure 4: Heterogeneous Effects of Demand Shocks on Automated Regulating & Control Instruments: Differentiated Industries

Notes: This figure displays the results of coefficient difference tests for imported regulating & control instruments across various goods sectors, along with 90% and 95% confidence intervals. Sectors are categorized as Differentiated, Reference, and Homogenous Goods according to Rauch (2001) (concorded to HS2-level). High and low Coefficient of Variation (CV) price sectors are defined by whether their HS2-level CV of prices is above or below the median from aggregated 2001-2025 China Customs data, following Cadot et al. (2013). The results show that the effect of demand shocks on regulating instruments are more significant in differentiated, reference goods sectors and sectors with high price dispersion.

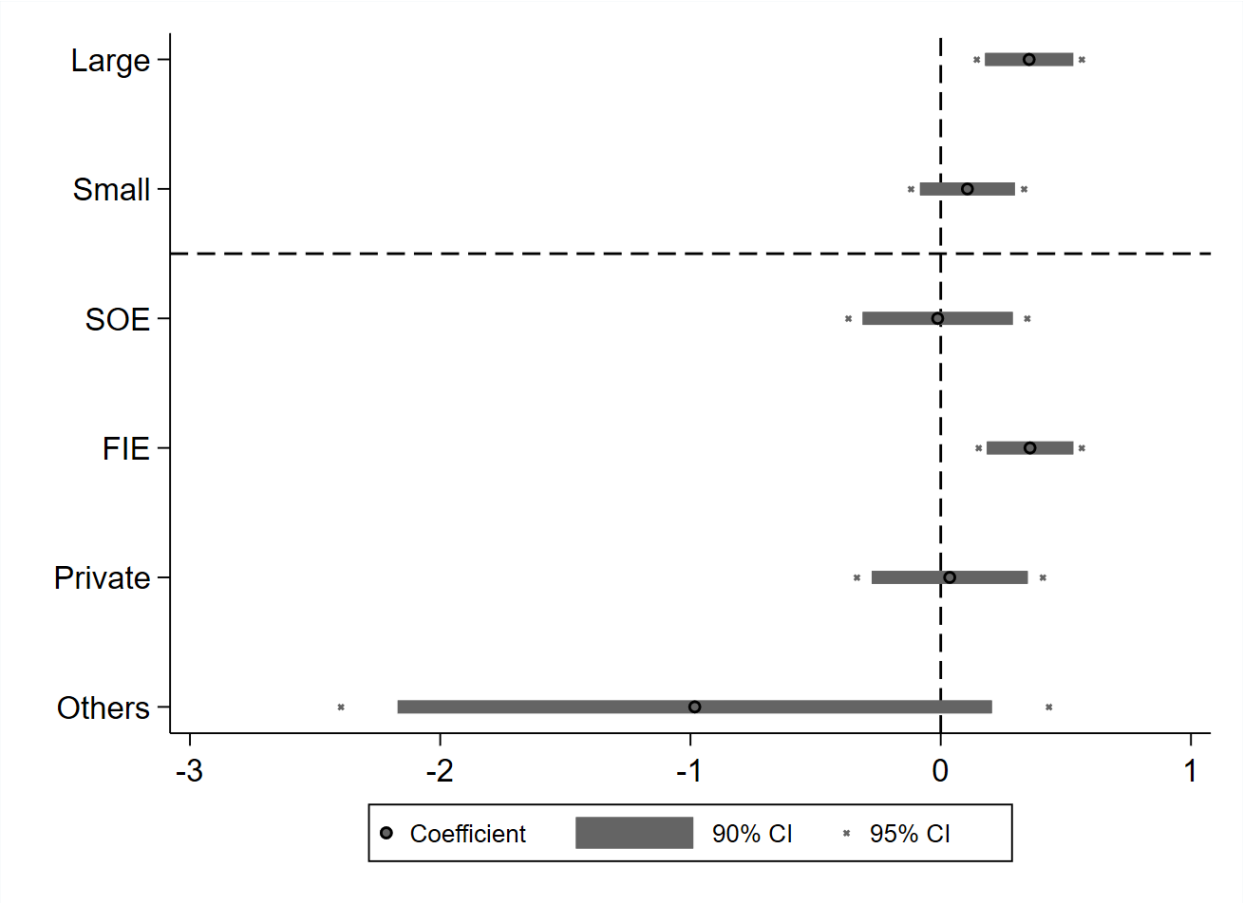


Figure 6: Heterogeneous Effects of Demand Shocks on Automated Regulating & Control Instruments: Firm Size and Ownership

Notes: This figure displays the results of coefficient difference tests for the impact of demand shocks on imported regulating and control instruments, presented with 90% and 95% confidence intervals. These tests are conducted across firm types: large vs. small firms (defined by initial export value from 2001-2015) and by ownership. Ownership categories include State-owned Enterprises (SOEs), Foreign-owned Enterprises (FOEs), Private firms, and Others (e.g., Collective firms). The sixth digit of the Chinese Customs Data firm ID shows firms' ownership upon registration. The result shows that the coefficient difference tests are significantly positive for large firms and FIEs, suggesting these firms are more responsive for demand shocks.

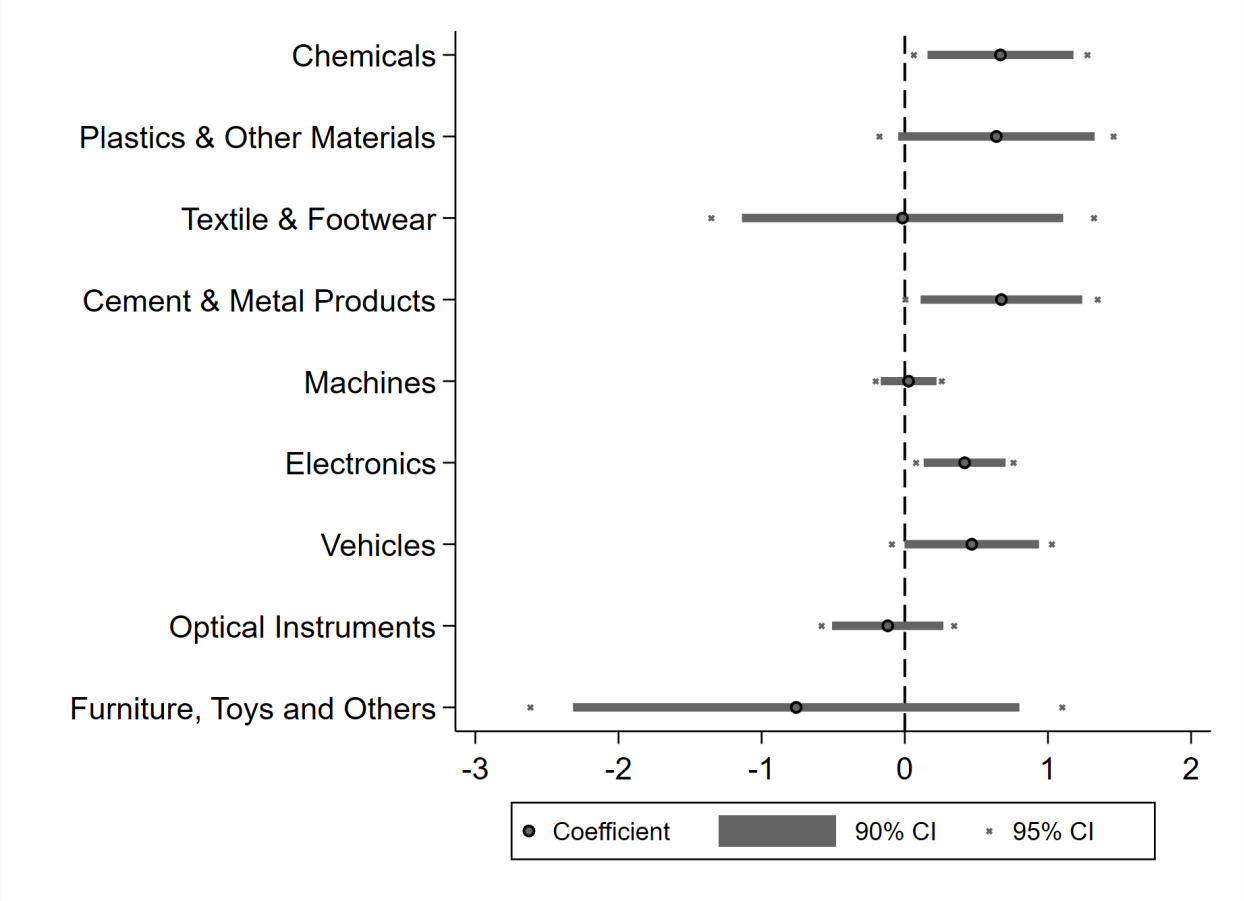
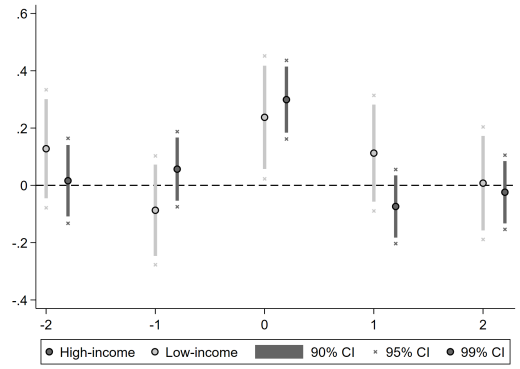
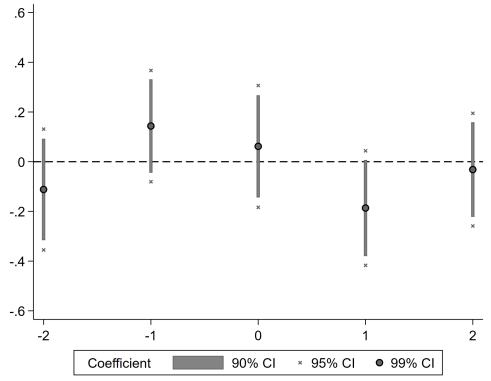


Figure 8: Heterogeneous Effects of Demand Shocks on Automated Regulating & Control Instruments

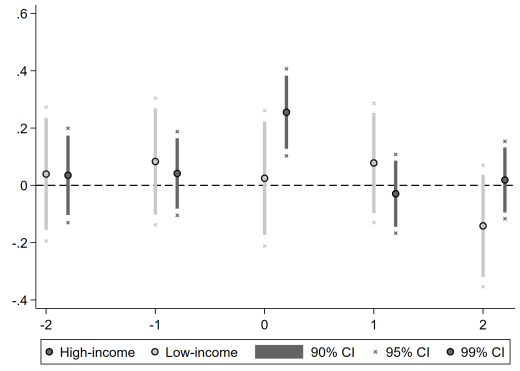
Notes: This figure illustrates the results of the coefficient difference test for each regression, displaying the 90% and 95% confidence intervals for imported regulating and control instruments across various aggregated industries. These industries are defined according to HS2 sectors: the Agricultural sector (HS codes 1-24) and the Mining sector (HS 25-27) were excluded from regression analysis due to an insufficient number of observations. Other sectors include Chemicals (HS 28-38); Plastics and Other Materials, which combines Plastic and Rubber (HS 39-40) with Leather, Wood, and Books (HS 41-49); Textile and Footwear, encompassing Textiles (HS 50-63) and Shoes, Hats, Umbrellas, and Fur (HS 64-67); Cement & Metal Products, comprising Cement, Ceramics, and Pearls (HS 68-70) along with Metals (HS 71-83); Machines (HS 84); Electronics (HS 95); Vehicles, covering Boats, Planes, and other Vehicles (HS 86-89); Optical Instruments (HS 90-92); and Furniture, Toys, and Others (HS 93-98). The results show that the coefficient difference tests are significantly positive in Chemicals, Cement and Metal Products and Electronics, consistent with the result that differentiated and refernces goods sectors showing more significant results.



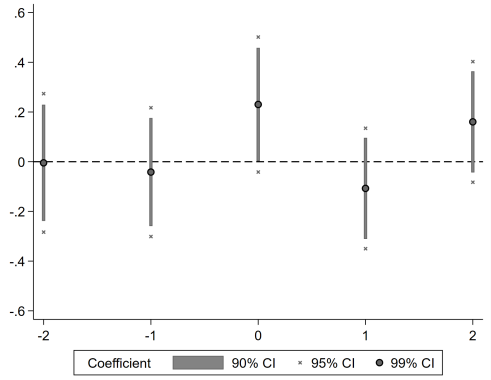
(a) Coefficients of Demand Shocks: Automation Machines



(b) Coefficient Difference Tests: Automation Machines



(c) Coefficients of Demand Shocks: Regulating Instruments



(d) Coefficient Difference Tests: Regulating Instruments

Figure 10: Coefficients of Demand Shocks and Coefficient Difference Tests for Imported Automated Machines and Regulating & Control Instruments

Notes: This figure presents the coefficients of high- and low- demand shocks and the results of the coefficient difference test, examining the dynamic effects of the demand shocks on imported automation machine and regulating & control instruments respectively, along with the 90% (as dots) and 95% (as bars) confidence intervals for imported automated machines. The contemporaneous effect is positive but not significant for automation imports, however is significantly positive for regulating & control instruments even when the sample size shrinks to half after including 2 leads and 2 lags of demand shocks.

Tables

Table 1: Share of Aggregate Machines in Total Imports (%)

Year	Share in total import (%)			Total import (tn. USD)
	Automation machines	Non-automated capital goods	Non-industrial machines	
2000	2.5	8.9	2.5	100.0
2001	3.3	10.1	2.7	124.3
2002	3.9	11.1	2.6	128.9
2003	3.8	11.1	2.3	187.8
2004	3.5	10.3	1.8	247.6
2005	3.0	9.0	1.7	278.3
2006	2.9	9.1	1.4	332.0
2007	2.9	8.2	1.2	428.6
2008	2.5	7.2	0.9	572.0
2009	2.8	8.2	1.0	534.2
2010	3.1	8.6	0.8	768.4
2011	3.0	8.1	0.7	1006.2
2012	3.0	6.4	0.8	1020.9
2013	2.8	6.0	0.8	1106.5
2014	3.1	6.6	0.9	1052.7
2015	3.2	6.7	1.0	921.0

Notes: This table presents the share of automation machines, non-automated machines and non-industrial machines in China's annual total import. The sample was restricted to ordinary trade, excluded trading companies and countries in Appendix A.2.

Table 2: Share of Specific Machines in Imports of Automation Machines (%)

Year	Share in imports of automation machines (%)							
		Other auto machines					Auto	
	Robot	Numerically controlled	Auto welding	Weaving & knitting	Other textile	Auto conveyor	machine tools	Regulating & control
2000	0.6	5.6	0.9	17.6	13.7	1.8	13.4	46.3
2001	0.7	6.6	1.5	13.4	16.6	2.1	12.7	46.4
2002	0.5	6.4	1.3	19.2	15.7	1.5	12.1	43.2
2003	0.9	6.9	0.8	17.1	16.0	1.8	12.2	44.5
2004	1.3	7.9	0.9	12.2	13.4	2.5	16.3	45.5
2005	1.2	8.5	1.1	10.7	9.4	2.4	15.6	51.2
2006	1.0	5.9	1.2	12.6	9.1	2.6	14.5	53.1
2007	0.9	7.8	1.0	13.0	9.5	1.9	14.6	51.3
2008	1.0	6.7	1.2	7.6	6.8	2.1	14.6	60.1
2009	1.5	7.0	0.9	5.3	4.5	2.6	15.5	62.6
2010	1.6	7.8	0.9	6.7	6.2	2.9	16.8	57.2
2011	2.2	7.8	0.8	5.7	7.1	2.9	17.5	56.0
2012	2.1	7.9	0.7	3.4	5.7	3.2	17.2	59.9
2013	1.9	7.2	0.6	3.9	5.9	3.1	14.7	62.6
2014	2.4	6.4	0.6	3.1	5.3	2.7	16.0	63.5
2015	2.3	6.3	0.6	2.8	4.5	3.3	15.2	65.0

Notes: This table presents the share of specific machines in imports of automation machines. The sample was restricted to ordinary trade, excluded trading companies and countries in Appendix Table A.2.

Table 3: The Impacts of Demand Shocks on Imports of Machines and Automated Machines

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Extensive		Intensive		Imputed capital stock		Customs-ASIF	
Dependent Variables	$\Delta\mathbf{1}$(Machine)	$\Delta\mathbf{1}$(Auto)	$\Delta \ln$ Machine	$\Delta \ln$ Auto	$\Delta \ln$ Machine	$\Delta \ln$ Auto	$\Delta \ln$ Auto	$\Delta \ln$ Auto
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft}$	0.017***	0.017***	0.225***	0.211***	0.075***	0.075***	0.258***	0.258***
	(0.005)	(0.005)	(0.031)	(0.045)	(0.012)	(0.015)	(0.048)	(0.048)
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft}$	0.005	0.003	0.149***	0.070	0.002	0.026	0.121*	0.121*
	(0.007)	(0.007)	(0.044)	(0.059)	(0.016)	(0.019)	(0.066)	(0.066)
$\Delta \ln XR^{M}_{ft}$	0.002	0.002	0.014	0.009	-0.020**	-0.025**	-0.015	-0.015
	(0.002)	(0.002)	(0.019)	(0.025)	(0.008)	(0.012)	(0.031)	(0.031)
$\Delta \ln XR^{X}_{ft}$	-0.001	0.000	0.006	0.006	-0.004	-0.002	0.034	0.034
	(0.002)	(0.002)	(0.022)	(0.029)	(0.008)	(0.010)	(0.034)	(0.034)
$\Delta \ln tar^{Machine}_{ft}$			0.304***		0.053***			
			(0.039)		(0.017)			
$\Delta \ln tar^{Auto}_{ft}$				0.052		-0.020	0.049	0.049
				(0.050)		(0.023)	(0.051)	(0.051)
$\Delta \ln \text{Wage}_{ct}$	-0.006*	-0.008**	0.019	0.055	0.068***	0.076***	0.075	0.076
	(0.003)	(0.003)	(0.029)	(0.042)	(0.014)	(0.018)	(0.046)	(0.046)
$s_{CHN,f,t-1} \times \Delta \ln \tilde{D}_{CHN,ft}$								0.079
								(0.077)
Observations	406,320	406,320	188,405	111,346	236,047	150,065	86,348	86,348
Adj. R^2	0.005	0.003	0.020	0.020	0.029	0.026	0.021	0.021
Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects				Industry-Province-Year				
$\beta_H - \beta_L$	0.013	0.014*	0.077	0.141**	0.073***	0.048**	0.137*	0.137*
s.e.$\beta_H - \beta_L$	(0.008)	(0.008)	(0.053)	(0.072)	(0.02)	(0.023)	(0.079)	(0.079)

Notes: This table examines the impact of demand shocks on firms' imports of both general machines and automation machines, using different constructions of the dependent variables, specifically through extensive margins, a perpetual inventory measure, intensive margin, and mid-point growth. Columns 1 and 2 examine the extensive margin, with the dependent variable being a dummy indicating the change in a firm's decision to import machines or automation machines. Columns 3 and 4 analyze the perpetual inventory measure, where the dependent variable is the log difference of the imputed stock of imported machines or automation machines, calculated using Brandt et al. (2012)'s method. Columns 5 and 6 focus on the intensive margin, using the log difference of imported machines or automation machines as the dependent variable. Finally, Columns 7 shows the impact of demand shocks on the log difference of firm imported automation machines in the Customs-ASIF merged sample, without controlling for the domestic shock. Column 8 further controls for the domestic demand shock. Standard errors, reported in parentheses, are clustered at the firm level. Standard errors (in parentheses) are clustered at the firm level.*** p<0.01, ** p<0.05, * p<0.1

Table 4: Accounting for Domestic Demand Shocks: Automated Machines

Dependent variables	(1) $\mathbf{1}$(Auto)	(2) $\mathbf{1}$(Auto)	(3) $\Delta \ln$ Auto	(4) $\Delta \ln$ Auto
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft}$	0.010	0.010	0.258***	0.258***
	(0.008)	(0.008)	(0.048)	(0.048)
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft}$	0.009	0.009	0.121*	0.121*
	(0.012)	(0.012)	(0.066)	(0.066)
$s_{CHN,f,t-1} \times \Delta \ln \tilde{D}_{CHN,ft}$		0.020		0.079
		(0.013)		(0.077)
Observations	115,496	115,496	86,348	86,348
Adj. R^2	0.053	0.053	0.021	0.021
Controls	Yes	Yes	Yes	Yes
Fixed effects		Industry-Province-Year		
$\beta_H - \beta_L$	0.001	0.001	0.137*	0.137*
s.e.$(\beta_H - \beta_L)$	(0.014)	(0.014)	(0.079)	(0.079)

Notes: This table examines the effect of demand shocks on firm imported automation machines accounting for domestic demand shocks. Column 1 and 2 present results through the extensive margins while column 3 and 4 discuss the intensive margin. Column 1 examines the impact of demand shocks on the changes of the dummy variable indicating whether firms import automation machines without controlling for domestic demand shock in the Customs-ASIF merged sample. Details for merging the Customs Data and the ASIF are reported in Appendix F. Column 2 further controls for adjusted domestic demand shocks. Adjusted domestic shocks ($s_{CHN,f,t-1} \times \Delta \ln \tilde{D}_{CHN,ft}$) are constructed in Equation 15, interacting firm-level domestic sales share ($s_{CHN,f,t-1}$) with the log change of the leave-one-province-out total sales ($\Delta \ln Y_{i(f),p(f),t}$). Domestic sales and total sales data come from the Annual Survey of Industrial Firms (ASIF). Panel A repeats the baseline regression in Table 3 using the Customs-ASIF merged sample controlling for firm-specific import and export exchange rates, firm-specific tariff and the log change of local wages, as detailed in Section 4. Panel B further controls for domestic demand shock. The results remain robust after accounting for domestic demand shocks. Standard errors (in parentheses) are clustered at the firm level.*** p<0.01, ** p<0.05, * p<0.1

Table 5: Robustness: The Impacts of Demand Shocks on Imports of Automated Machines

	(1)	(2)	(3)	(4)	(5)	(6)
	Mid-point	t-2	Bootstrap	Trading Firms	Intensive	
Dependent variables	Automid	$\Delta \ln$ Auto	$\Delta \ln$ Auto	$\Delta \ln$ Auto	$\Delta \ln$ Nonauto	$\Delta \ln$ Notind
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft}$	0.152***	0.209***	0.211***	-0.139	0.208***	0.190
	(0.025)	(0.048)	(0.050)	(0.132)	(0.034)	(0.126)
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft}$	0.034	0.071	0.070	-0.136	0.155***	0.121
	(0.036)	(0.067)	(0.063)	(0.105)	(0.051)	(0.127)
Observations	162,814	79,158	111,346	21,184	160,571	18,347
Adj. R^2	0.038	0.023	0.020	0.014	0.019	0.025
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects			Industry-Province-Year			
$\beta_H - \beta_L$	0.118***	0.138*	0.141*	-0.002	0.053	0.07
s.e.$(\beta_H - \beta_L)$	(0.043)	(0.08)	(0.076)	(0.161)	(0.06)	(0.175)

Notes: This table shows the robustness test for the impact of demand shocks on firm imported automation machines. Column 1 takes the extensive margin into consideration by following Aghion et al. (2024) to construct the mid-point growth rate for both the explanatory variables and the outcome variables. Column 2 uses demand shocks in t-2 period. Column 3 reports bootstrap standard errors. The coefficient difference tests are significantly positive in column 1 to 3. Column 4 reports robustness test for trading companies. No significant effects were found for trading companies as expected, as many trading companies are not involved in direct production. Colum 5 and 6 reports the intensive margin results for non automation machines and non-industrial machines. Either has significant coefficient test result as expected. It shows the impact of demand shocks on machines is mainly driven by automation machines.

Table 6: Robustness: The Impacts of Demand Shocks on Imports of Specific Automated Machines

	(1)	(2)	(3)	(4)
Dependent variables	$\Delta \ln$ Robot	$\Delta \ln$ Auto machine	$\Delta \ln$ Auto machtool	$\Delta \ln$ Regulate
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft}$	0.124	0.153	0.200**	0.216***
	(0.434)	(0.132)	(0.081)	(0.050)
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft}$	0.186	0.034	0.090	-0.034
	(0.545)	(0.138)	(0.131)	(0.065)
Observations	556	14,841	40,150	83,517
Adj. R^2	0.051	0.028	0.020	0.018
Controls	Yes	Yes	Yes	Yes
Fixed Effects		Industry-Province-Year		
$\beta_H - \beta_L$	-0.062	0.119	0.110	0.250***
s.e.$(\beta_H - \beta_L)$	(0.686)	(0.184)	(0.151)	(0.080)

Notes: This table present results for specific imported automation machines including robot in column 1, automation machines, in column 2, automation machine tools in column 3, regulating and control instruments in column 4. The coefficient tests are only significantly positive for regulating and control instruments showing that the impact of demand shocks on automation machines are primarily driven by regulating and control instruments.

Table 7: Heterogeneity: The Impacts of Demand Shocks on Imports of Regulating and Control Instruments - Sector Characteristics

Dependent Variables	(1) *Rauch* $\Delta \ln$ Regulate	(2) *CV_price* $\Delta \ln$ Regulate
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft} \times diff$	0.196***	
	(0.052)	
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft} \times ref$	0.429**	
	(0.188)	
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft} \times homo$	1.762*	
	(0.930)	
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft} \times diff$	-0.014	
	(0.068)	
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft} \times ref$	-0.301*	
	(0.175)	
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft} \times homo$	2.163**	
	(1.027)	
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft} \times high_cv_price$		0.209***
		(0.053)
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft} \times low_cv_price$		0.288*
		(0.155)
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft} \times high_cv_price$		-0.049
		(0.066)
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft} \times low_cv_price$		0.096
		(0.236)
Observations	83,517	83,517
Adj. R^2	0.018	0.018
Controls	Yes	Yes
Fixed Effects	Industry-Province-Year	
$diff(\beta_H - \beta_L)$	0.21**	
$diff_se(\beta_H - \beta_L)$	(0.084)	
$ref(\beta_H - \beta_L)$	0.73***	
$ref_se(\beta_H - \beta_L)$	(0.25)	
$homo(\beta_H - \beta_L)$	-0.401	
$homo_se(\beta_H - \beta_L)$	(1.355)	
$high_cv_price(\beta_H - \beta_L)$		0.258***
$high_cv_price_se(\beta_H - \beta_L)$		(0.083)
$low_cv_price(\beta_H - \beta_L)$		0.192
$low_cv_price_se(\beta_H - \beta_L)$		(0.271)

Notes: This table shows the results for imported regulating & control instruments across various goods sectors. Sectors are categorized as Differentiated, Reference, and Homogenous Goods according to Rauch (2001) (concorded to HS2-level). High and low Coefficient of Variation (CV) price sectors are defined by whether their HS2-level CV of prices is above or below the median from aggregated 2001-2025 China Customs data, following Cadot et al. (2013). The results shows that the coefficient difference test is significant for differentiated and reference goods sectors, but not significant for homogeneous goods sectors. And it is significant for sectors with high coefficient of variation of price, the other indicator of product differentiation.

Table 8: Heterogeneity: The Impacts of Demand Shocks on Imports of Regulating and Control Instruments - Firm Characteristics

	(1)	(2)
	Size	Ownership
Dependent Variables	$\Delta \ln$ Regulate	$\Delta \ln$ Regulate
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft} \times highexpv$	0.291***	
	(0.064)	
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft} \times lowexpv$	0.108	
	(0.069)	
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft} \times highexpv$	-0.062	
	(0.086)	
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft} \times lowexpv$	0.002	
	(0.094)	
$highexpv$	-0.015	
	(0.011)	
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft} \times soe$		-0.013
		(0.134)
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft} \times fie$		0.267***
		(0.057)
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft} \times private$		0.088
		(0.122)
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft} \times colother$		-0.333
		(0.574)
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft} \times soe$		-0.001
		(0.122)
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft} \times fie$		-0.089
		(0.091)
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft} \times private$		0.052
		(0.140)
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft} \times colother$		0.649
		(0.413)
soe		0.058
		(0.065)
fie		0.179***
		(0.062)
$private$		0.182***
		(0.064)
Observations	83,517	83,517
Adj. R^2	0.018	0.019
Controls	Yes	Yes
Fixed Effects	Industry-Province-Year	
$highexpv(\beta_H - \beta_L)$	0.353***	
$highexpv_se(\beta_H - \beta_L)$	(0.107)	
$lowexpv(\beta_H - \beta_L)$	0.107	
$lowexpv_se(\beta_H - \beta_L)$	(0.115)	
$soe(\beta_H - \beta_L)$		-0.012
$soe_se(\beta_H - \beta_L)$		(0.182)
$fie(\beta_H - \beta_L)$		0.357***
$fie_se(\beta_H - \beta_L)$		(0.105)
$private(\beta_H - \beta_L)$		0.036
$private_se(\beta_H - \beta_L)$		(0.19)
$colother(\beta_H - \beta_L)$		-0.983
$colother_se(\beta_H - \beta_L)$		(0.722)

Notes: This table presents results for the impact of demand shocks on imported regulating instruments for larger and smaller firms and for firms with different ownerships (SOE, FIE, Private and Others). The coefficient difference test is significantly positive for larger firms, and for FIE only.

Table 9: Price and Quality Changes

	(1)	(2)
Dependent variables	$\Delta \ln$ Price	ΔQuality
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft}$	0.025***	0.054***
	(0.007)	(0.011)
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft}$	0.001	0.033**
	(0.010)	(0.015)
Observations	2,390,069	2,388,757
Adj. R^2	0.028	0.009
Control	Yes	Yes
Fixed effects	Industry-Province-Year, Product-Year	
$\beta_H - \beta_L$	0.025**	0.021
s.e.$(\beta_H - \beta_L)$	(0.012)	(0.018)

Notes: This table presents results for price and quality changes. The dependent variables are log change of export prices (column 1), and firm-HS 6 product-year level export quality (column 2) estimated following Khandelwal et al. (2013) in equation (18). Control variables include firm-specific import and export exchange rates, firm-specific tariff and the log change of local wages, as detailed in Section 4. The number of observations changes across columns because each first-difference regression retains only firms that have price and export quality data for two consecutive years. We include industry-province-year fixed effects and HS 6-digit product-year fixed effects. Standard errors (in parentheses) are clustered at the firm level. *** p<0.01, ** p<0.05, * p<0.1

Appendix

A Data Appendix

A.1 Classification of Machines

The appendix is adapted from the working paper version of Acemoglu and Restrepo (2022). We added detailed description for "regulating & control instruments" as this category turns out to play a key role in the overall adoption of automation equipment.

We focus on imported machines and instruments, defined as products whose HS two-digit code is given by 82 (Tools), 84 (Mechanical machinery and appliances), 85 (Electrical machinery and equipment), 87 (Tractors and work trucks), and 90 (Instruments and apparatus). We divided all goods into three categories: automation equipment, non-automated capital goods, and non-industrial machines. To measure these goods as accurately as possible, we defined the categories using the HS 1996, HS 2002, HS 2007 and HS 2012 classifications respectively. The three categories used in the paper are defined as follows:

The first set is automation machines: industrial robots (*Robot*), numerically controlled machines, automatic machine tools (*Auto machtools*), automatic welding machines, weaving and knitting machines, other textile dedicated machinery, automatic conveyors, regulating and control instruments (*Regulate*). Among them, the combination of numerically controlled machines, automatic welding machines, weaving and knitting machines, other textile dedicated machinery, and automatic conveyors is automated machines (*Auto machines*).

The second set comprises non-automated capital goods used for similar industrial tasks. It is further divided into machines (*Non auto machine*) and tools (*Non auto tools*). Machines include heavy capital goods (including furnaces, ovens, and electrical motors), food manufacturing (machines used for brewing and baking in industrial contexts), not-numerically controlled machines, and manual welding machines. Tools include tools for industrial work, manual machine tools, and tools for transferring materials.

The third set of non-industrial machines or tools includes laundry machines, vending machines, agricultural machinery, and computers. These machines are not used for industrial production.

Now we describe the details of the identification for each specific category:

- Industrial robots: This category includes industrial robots. It is defined by the six-digit HS code 847950. This category was introduced to the HS-1996 classification.

- Numerically controlled machines : For a wide class of metal-working machines (lathes, milling machines), the HS classification distinguishes "numerically controlled" vintages from "other than numerically controlled." Based on this distinction we create two separate categories: numerically controlled machines and not-numerically controlled vintages.

- Machine tools: For a wide class of machine tools, the HS classification distinguishes those that are for "working with hands" from the rest. Based on this distinction we create two separate categories: automatic machine tools and manual machine tools.

- Tools for industrial work: This category includes tools (not machines or machine tools) used in industrial applications.

- Welding machines: For welding machines, the HS classification distinguishes those that are automatic from those that are not. Based on this distinction we create two separate categories: automatic welding machines and manual welding machines.

- Weaving and knitting machines: This category includes weaving and knitting machines used in the textile industry, excluding textile appliances for household use (such as sewing machines for household use). We grouped the remaining dedicated machinery used in textiles into other textile dedicated machinery.

- Conveyors: For conveyors, the HS classification distinguishes those that are "continuous-action" and therefore automatic from other machinery that transfer or move materials with human operation (like work trucks). Based on this distinction we create two separate categories: automatic conveyors and tools for transferring materials.

- Regulating and control instruments: This category includes instruments used for control applications in manufacturing, for instance, instruments and apparatus used for measuring, checking, analyzing, regulating, or controlling various physical or chemical

properties. These instruments are likely used in industries that rely on precise control and monitoring. The subcategories include:

  - Measuring instruments: These measure various physical quantities like temperature, pressure, and the flow or level (HS codes 9025, 9026).
  - Analysis instruments: These analyze the composition or properties of materials (HS codes 9027).
  - Meters and counters: These measure specific quantities like electricity, gas, revolutions or production (HS codes 9028, 9029).
  - Electrical instruments: These measure or analyze electrical properties such as voltage, current, resistance or power (HS codes 9030).
  - Other Measuring or checking instruments, appliances and machines (HS codes 9031).
  - Automatic regulating or controlling instruments and apparatus (HS codes 9032).

- Heavy capital goods: This category includes heavy capital goods with applications in industry that are not related to automation. These more traditional types of capital goods include boilers, furnaces, ovens and electric motors.
- Food manufacturing: This category includes machines used for baking, brewing, and preparing food and beverages in an industrial context and excluding home appliances.
- Vending machines: This category includes vending machines and their parts.
- Laundry machines: This category includes laundry machines and their parts.
- Agricultural machinery: This category includes agricultural machinery and tractors.
- Computers: This category includes computers and their parts (not software).

A.2 Appendix for countries or regions dropped from the Customs Data

The countries or regions dropped from the Chinese Customs Data fall into the following categories:

Countries or regions included in BACI but without income classification: British Virgin Islands, Cook Islands, Curaçao, Gibraltar, Mayotte (Overseas France), Montenegro, Montserrat, Nauru, Netherlands Antilles (...2010), Norfolk Islands, Saint Pierre and Miquelon, Serbia, Serbia and Montenegro (...2005), South Sudan, Turks and Caicos Islands, Tuvalu, Wallis and Futuna Islands.

Countries or regions not included in BACI but with income classification: Faroe Islands, Liechtenstein, Monaco, Puerto Rico.

Countries or regions not included in BACI and without income classification: Bonaire, Cambier Islands, Canary Islands, Ceuta, Countries (regions) unknown, Countries of Neutral Package, French Guiana, Guadeloupe, Marquesas Islands, Martinique, Melilla, Other African countries (regions), Other Asian countries (regions), Other European countries (regions), Other Latin American countries (regions), Other North American countries (regions), Other Oceanian countries (regions), Reunion, Saba, Saint Martin Islands, Society Islands, Tuamotu Islands, Tubai Islands, UN and other international organizations, Vatican City State, Western Sahara.

A.3 Appendix for Merging the Customs Data and the Annual Survey of Industrial Firms

China's Annual Survey of Industrial Firms (ASIF) comes from the National Bureau of Statistics (NBS) in China, which includes various variables reflecting production, sales, and financial conditions from the balance sheets, income statements, and cash flow statements of these enterprises. This includes variables such as sales, output, number of employees, and revenue. Since 1998, the NBS has divided industrial statistics into two categories: large-scale and small-scale enterprises. "Large-scale industrial enterprises" from 1998 to 2006 referred to all state-owned and non-state-owned industrial legal entities with annual main business income of 5 million yuan or more; from 2007 to 2010, the statistical scope was adjusted to industrial legal entities with annual main business income of 5 million yuan or more; and

from 2011 till now, the statistical scope is industrial legal entities with annual main business income of 20 million yuan or more. In this study, we use the ASIF data to obtain total sales for each firm.

We refer to Brandt et al. (2012) to link the the firms in ASIF from 1998 to 2013. Then, we match the Annual Survey of Industrial Firms with the Customs Data firms database using firm names, postal codes, and phone numbers. We go through two steps to match the two dataset following Yu (2015). First, we match firms with exact firm names in the ASIF and the Customs Data. Second, we match firms with identical postal codes and the last seven digits of their phone numbers. The rationale is that firms should have different and unique phone numbers within a postal district. The majority of firms are matched in the first step. The matching results are reported in the Appendix Table A1. The merged dataset covers 174,652 firms (26.1% of Customs Data firms, 20.2% of ASIF firms) and 77.5% of export value in the filtered ASIF data. The matching quality is comparable to Yu (2015) who matched 27% of the 286,819 firms in the product-level trade data set from 2000-2006 and approximately 17% of the 438,146 valid firms in the firm-level production data set and around 53% of exports in terms of export value.

Table A1: Number of Firms in the Customs Data, the Annual Survey of Industrial Firms and the Merged Data

year	Customs Data	ASIF	Filtered ASIF	Matched Firms	Share of Matched Firms in Customs	Share of Matched Firms in ASIF	Share of Matched Firms in Filtered ASIF
2001	89,660	171,256	141,087	28,255	31.5	16.5	16.8
2002	104,246	181,557	148,349	33,083	31.7	18.2	18.8
2003	124,299	196,222	169,648	39,033	31.4	19.9	20.1
2004	153,780	279,092	250,725	56,501	36.7	20.2	20.3
2005	179,663	270,043	244,331	58,913	32.8	21.8	21.7
2006	208,425	301,961	274,202	65,666	31.5	21.7	21.8
2007	236,498	336,768	308,018	73,526	31.1	21.8	21.6
2008	251,962	412,212	381,880	86,259	34.2	20.9	20.6
2009	262,976	366,130	341,150	76,531	29.1	20.9	20.1
2010	285,686	442,539	421,934	89,353	31.3	20.2	21.0
2011	310,866	302,593	294,311	73,931	23.8	24.4	24.6
2012	328,826	324,604	300,435	77,439	23.6	23.9	24.9
2013	344,983	344,875	332,441	80,649	23.4	23.4	23.7
all years	667,925	866,739	814,349	174,652	26.1	20.2	21.0

Notes: The table show the number of firms in the Chinese Customs Data, ASIF, filtered ASIF and the merged dataset. The merged datasets cover 36.4% of output in current prices and 37.0% of sales and 77.5% of exports and 35.4% of employment in Filtered ASIF. In all ASIF, the merged dataset covers 37.3% of output in current prices, 37.9% of sales, 78.1% of exports and 35.8% employment. Our matching quality is comparable to Yu (2015) who matched 27% of the 286,819 firms in the product-level trade data set from 2000-2006 and approximately 17% of the 438,146 valid firms in the firm-level production data set and around 53% of exports in terms of export value.

B Appendix Tables

Table B1: Robustness: The Impacts of Demand Shocks on Imports of Regulating and Control Instruments

Dependent Variables	(1) Customs-ASIF $\Delta \ln$ Regulate	(2) Customs-ASIF $\Delta \ln$ Regulate	(3) t-2 $\Delta \ln$ Regulate	(4) boostrap $\Delta \ln$ Regulate
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft}$	0.252***	0.252***	0.217***	0.216***
	(0.055)	(0.055)	(0.053)	(0.053)
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft}$	-0.012	-0.013	0.005	-0.034
	(0.072)	(0.072)	(0.072)	(0.071)
$s_{CHN,f,t-1} \times \Delta \ln \tilde{D}_{CHN,ft}$		0.107		
		(0.087)		
Observations	63,626	63,626	59,279	83,517
Adj. R^2	0.019	0.019	0.023	0.018
Controls	Yes	Yes	Yes	Yes
Fixed Effects		Industry-Province-Year		
$\beta_H - \beta_L$	0.265***	0.265***	0.212**	0.25***
s.e.$(\beta_H - \beta_L)$	(0.088)	(0.088)	(0.089)	(0.081)

Notes: This table examines the impact of demand shocks on imports of regulation and control instruments through intensive margins, t-2 period shock and bootstrapped standard errors. The dependent variable in column 1 presents baseline result in Customs-ASIF merged data, column 2 further control for domestic demand shock. Column 3 report results for t-2 period demand shock. Column 4 reports bootstrapped standard errors.

Table B2: Placebo Test: Impact of Demand Shocks on Non-automated Capital Goods and Non-industrial Machines

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variables	$\Delta \ln$ Non auto capital	$\Delta \ln$ Non auto machine	$\Delta \ln$ Non auto tool	$\Delta \ln$ Non-industrial machines	$\Delta \ln$ Other non-industrial machines	$\Delta \ln$ Computer
$s_{H,f} \times \Delta \ln \tilde{D}_{H,ft}$	0.208***	0.207***	0.147***	0.190	0.421	0.139
	(0.034)	(0.037)	(0.055)	(0.126)	(0.298)	(0.146)
$s_{L,f} \times \Delta \ln \tilde{D}_{L,ft}$	0.155***	0.156***	0.101	0.121	0.229	0.028
	(0.051)	(0.054)	(0.091)	(0.127)	(0.225)	(0.156)
Observations	160,571	144,932	59,634	18,347	4,788	13,013
Adj. R^2	0.019	0.020	0.018	0.025	0.057	0.014
Control	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects			Industry-Province-Year			
$\beta_H - \beta_L$	0.053	0.051	0.045	0.07	0.193	0.111
s.e.$(\beta_H - \beta_L)$	(0.06)	(0.064)	(0.104)	(0.175)	(0.358)	(0.21)

Notes: This table presents our placebo test for imported non-automated capital goods and non-industrial machines. The dependent variables in column 1-3 are the log changes in imported non-automated capital goods (column 1), including non-automated machines (column 2) and non-automated tools (column 3). The dependent variables in column 4-6 are the log changes in imported non-industrial machines (column 4), including computer (column 5) and other non-industrial machines (column 6). Detailed descriptions of the dependent variables are provided in Appendix A. The control variables include firm-specific import and export exchange rates, firm-specific tariff and the log change of local wages, as detailed in Section 4. The number of observations changes across columns because each first-difference regression retains only firms that import the corresponding machines for two consecutive years. Each regression controls for industry-province-year fixed effects, where industry (time-invariant) is defined by the largest HS 2-digit industry that a firm exports from 2001 to 2015. Coefficients of demand shocks are insignificant as expected. Standard errors (in parentheses) are clustered at the firm level. *** p<0.01, ** p<0.05, * p<0.1

Table B3: Placebo Test of Trading Companies: Impact of Demand Shocks on Imported Automated Machines

	(1)	(2)	(3)	(4)	(5)
Dependent variables	$\Delta \ln$ Auto	$\Delta \ln$ Robot	$\Delta \ln$ Auto machine	$\Delta \ln$ Auto machtool	$\Delta \ln$ Regulate
$s_{H,f} \times \Delta \ln \tilde{D}_{H,ft}$	-0.139	1.085	-0.051	0.075	-0.055
	(0.132)	(0.903)	(0.320)	(0.225)	(0.145)
$s_{L,f} \times \Delta \ln \tilde{D}_{L,ft}$	-0.136	0.679	-0.068	-0.106	-0.012
	(0.105)	(2.206)	(0.262)	(0.270)	(0.110)
Observations	21,184	95	3,787	6,047	16,513
Adj. R^2	0.014	0.050	0.050	0.031	0.011
Control	Yes	Yes	Yes	Yes	Yes
Fixed effects			Industry-Province-Year		
$\beta_H - \beta_L$	-0.002	0.406	0.016	0.181	-0.043
s.e.$(\beta_H - \beta_L)$	(0.161)	(2.371)	(0.413)	(0.345)	(0.174)

Notes: This table presents our placebo test result for trading companies. The dependent variables are the log changes in imported automated machines (column 1), including robots (column 2), other automated machines (column 3), and automated machine tools (column 4) and regulating and control instruments (column 5). Detailed descriptions of the dependent variables are provided in Appendix A. The control variables include firm-specific import and export exchange rates, firm-specific tariff and the log change of local wages, as detailed in Section 4. The number of observations changes across columns because each first-difference regression retains only firms that import the corresponding machines for two consecutive years. Each regression controls for industry-province-year fixed effects, where industry (time-invariant) is defined by the largest HS 2-digit industry that a firm exports from 2001 to 2015. Coefficients of demand shocks are insignificant as expected. Standard errors (in parentheses) are clustered at the firm level. *** p<0.01, ** p<0.05, * p<0.1

Table B4: Placebo Test of Trading Companies: Impact of Demand Shocks on Imported Non-automated Capital Goods and Non-industrial Machines

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variables	$\Delta \ln$ Non auto capitals	$\Delta \ln$ Non auto machine	$\Delta \ln$ Non auto tools	$\Delta \ln$ Non-industrial machines	$\Delta \ln$ Other non-industrial machines	$\Delta \ln$ Computer
$s_{H,f} \times \Delta \ln \tilde{D}_{H,ft}$	-0.115	-0.067	-0.089	0.048	0.333	-0.260
	(0.101)	(0.106)	(0.208)	(0.267)	(0.420)	(0.328)
$s_{L,f} \times \Delta \ln \tilde{D}_{L,ft}$	-0.045	-0.002	-0.042	-0.185	-0.136	0.335
	(0.093)	(0.098)	(0.206)	(0.247)	(0.337)	(0.280)
Observations	31,321	28,814	9,712	5,480	1,767	3,663
Adj. R^2	0.020	0.019	0.016	0.037	0.015	0.035
Control	Yes	Yes	Yes	Yes	Yes	Yes
HS2-Prov-Year	Yes	Yes	Yes	Yes	Yes	Yes
$\beta_H - \beta_L$	-0.07	-0.065	-0.047	0.234	0.469	-0.595
s.e.$(\beta_H - \beta_L)$	(0.14)	(0.148)	(0.279)	(0.359)	(0.508)	(0.427)

Notes: This table presents our placebo test for trading companies on imported non-automated machines and non-industrial machines. The dependent variables in column 1-3 are the log changes in imported non-automated capital goods (column 1), including non-automated machines (column 2) and non-automated tools (column 3). The dependent variables in column 4-6 are the log changes in imported non-industrial machines (column 1), including computer (column 3) and other non-industrial machines (column 2). Detailed descriptions of the dependent variables are provided in Appendix A. The control variables include firm-specific import and export exchange rates, firm-specific tariff and the log change of local wages, as detailed in Section 4. The number of observations changes across columns because each first-difference regression retains only firms that import the corresponding machines for two consecutive years. Each regression controls for industry-province-year fixed effects, where industry (time-invariant) is defined by the largest HS 2-digit industry that a firm exports from 2001 to 2015. Coefficients of demand shocks are insignificant as expected. Standard errors (in parentheses) are clustered at the firm level. *** p<0.01, ** p<0.05, * p<0.1

Table B5: Robustness Test for Demand Shocks from $t-2$ Period

	(1)	(2)	(3)	(4)	(5)
Dependent variables	$\Delta \ln$ Auto	$\Delta \ln$ Robot	$\Delta \ln$ Auto machine	$\Delta \ln$ Auto machtool	$\Delta \ln$ Regulate
$s_{H,f} \times \Delta \ln \tilde{D}_{H,ft}$	0.209***	1.118*	0.045	0.217**	0.217***
	(0.048)	(0.572)	(0.147)	(0.096)	(0.053)
$s_{L,f} \times \Delta \ln \tilde{D}_{L,ft}$	0.071	-0.547	0.022	0.303*	0.005
	(0.067)	(0.784)	(0.195)	(0.177)	(0.072)
Observations	79,158	225	8,505	26,869	59,279
Adj. R^2	0.023	0.108	0.038	0.020	0.023
Control	Yes	Yes	Yes	Yes	Yes
Fixed effects			Industry-Province-Year		
$\beta_H - \beta_L$	0.138*	1.665**	0.023	-0.086	0.212**
s.e.$(\beta_H - \beta_L)$	(0.08)	(0.818)	(0.233)	(0.195)	(0.089)

Notes: This table shows results for $t-2$ share when constructing the demand shocks. The dependent variables are the log changes in imported automated machines (column 1), including robots (column 2), other automated machines (column 3), and automated machine tools (column 4) and regulating and control instruments (column 5). Detailed descriptions of the dependent variables are provided in Appendix A. The control variables include firm-specific import and export exchange rates, firm-specific tariff and the log change of local wages, as detailed in Section 4. The number of observations changes across columns because each first-difference regression retains only firms that import the corresponding machines for two consecutive years. Each regression controls for industry-province-year fixed effects, where industry (time-invariant) is defined by the largest HS 2-digit industry that a firm exports from 2001 to 2015. The results remain robust. Standard errors (in parentheses) are clustered at the firm level. *** p<0.01, ** p<0.05, * p<0.1

Table B6: Robustness Test for Bootstrapped Standard Errors

	(1)	(2)	(3)	(4)	(5)
Dependent variables	$\Delta \ln$ Auto	$\Delta \ln$ Robot	$\Delta \ln$ Auto machine	$\Delta \ln$ Auto machtool	$\Delta \ln$ Regulate
$s_{H,f} \times \Delta \ln \tilde{D}_{H,ft}$	0.209***	1.118**	0.045	0.217**	0.217***
	(0.049)	(0.561)	(0.150)	(0.097)	(0.054)
$s_{L,f} \times \Delta \ln \tilde{D}_{L,ft}$	0.071	-0.547	0.022	0.303*	0.005
	(0.068)	(0.913)	(0.178)	(0.175)	(0.074)
Observations	79,158	225	8,505	26,869	59,279
Adj. R^2	0.023	0.108	0.038	0.020	0.023
Control	Yes	Yes	Yes	Yes	Yes
Fixed effects			Industry-Province-Year		
$\beta_H - \beta_L$	0.138*	1.665*	0.023	-0.086	0.212**
s.e.$(\beta_H - \beta_L)$	(0.084)	(0.992)	(0.233)	(0.202)	(0.086)

Notes: This table reports bootstrapped standard errors (in parentheses) for our baseline regression in Table 3. The dependent variables are the log changes in imported automated machines (column 1), including robots (column 2), other automated machines (column 3), and automated machine tools (column 4) and regulating and control instruments (column 5). Detailed descriptions of the dependent variables are provided in Appendix A. The control variables include firm-specific import and export exchange rates, firm-specific tariff and the log change of local wages, as detailed in Section 4. The number of observations changes across columns because each first-difference regression retains only firms that import the corresponding machines for two consecutive years. Each regression controls for industry-province-year fixed effects, where industry (time-invariant) is defined by the largest HS 2-digit industry that a firm exports from 2001 to 2015. *** p<0.01, ** p<0.05, * p<0.1

Table B7: Robustness: The Impacts of Demand Shocks on Imports of Regulating and Control Instruments

Category	(1) Chemicals	(2) Plastics & Others	(3) Textile & Footwear	(4) Cement & Metal	(5) Machines	(6) Electronics	(7) Vehicles	(8) Optical Instruments	(9) Furniture & Others
Dependent Variables	$\Delta \ln$ Regulate	$\Delta \ln$ Regulate	$\Delta \ln$ Regulate	$\Delta \ln$ Regulate	$\Delta \ln$ Regulate	$\Delta \ln$ Regulate	$\Delta \ln$ Regulate	$\Delta \ln$ Regulate	$\Delta \ln$ Regulate
$s_{H,f,t-1} \times \Delta \ln \tilde{D}_{H,ft}$	0.432*	0.524*	0.228	0.512***	0.085	0.276***	0.131	0.121	0.259
	(0.234)	(0.306)	(0.410)	(0.185)	(0.074)	(0.101)	(0.198)	(0.151)	(0.539)
$s_{L,f,t-1} \times \Delta \ln \tilde{D}_{L,ft}$	-0.236	-0.115	0.244	-0.163	0.058	-0.142	-0.337*	0.240	1.018
	(0.222)	(0.300)	(0.542)	(0.304)	(0.096)	(0.145)	(0.204)	(0.190)	(0.785)
Observations	3,887	4,398	2,086	5,957	26,735	23,643	5,349	10,085	1,365
Adj. R^2	0.002	0.002	-0.011	0.022	0.012	0.015	0.011	0.011	0.016
Fixed Effects	Industry-Province-Year								
$\beta_H - \beta_L$	0.669**	0.64	-0.016	0.675**	0.027	0.418**	0.468	-0.119	-0.758
s.e.$(\beta_H - \beta_L)$	(0.309)	(0.417)	(0.681)	(0.343)	(0.118)	(0.174)	(0.285)	(0.236)	(0.946)

Notes:This table illustrates the results for imported regulating and control instruments across various aggregated industries. These industries are defined according to HS2 sectors: the Agricultural sector (HS codes 1-24) and the Mining sector (HS 25-27) were excluded from regression analysis due to an insufficient number of observations. Other sectors include Chemicals (HS 28-38); Plastics and Other Materials, which combines Plastic and Rubber (HS 39-40) with Leather, Wood, and Books (HS 41-49); Textile and Footwear, encompassing Textiles (HS 50-63) and Shoes, Hats, Umbrellas, and Fur (HS 64-67); Cement Metal Products, comprising Cement, Ceramics, and Pearls (HS 68-70) along with Metals (HS 71-83); Machines (HS 84); Electronics (HS 95); Vehicles, covering Boats, Planes, and other Vehicles (HS 86-89); Optical Instruments (HS 90-92); and Furniture, Toys, and Others (HS 93-98). The coefficient difference tests are significant for Chemicals, Cement and Metal Products, Electronics Porducts, consistent with the result for Rauch industries which shows the effects are significant for reference goods and differentiated goods.